Agentic Systems Patterns

A practical reference for modern agent architecture: goals, loops, tools, skills, memory, protocols, multi-agent systems, and production runtimes.

Agentic Systems Patterns

This book is a practical reference for designing, testing, and operating agentic systems.

The catalog favors patterns that help engineers build real systems: control loops, goals, state, tools, skills, protocols, memory, multi-agent coordination, systems architecture, runtime observability, and evaluation.

Runnable examples live in the repository root. Pattern chapters link back to the relevant source folder when code is available. Systems Architecture chapters explain how the patterns fit together into deployable systems.

Publishing

The public site is designed for GitHub Pages at:

```
https://gturitto.github.io/Agentic-Systems-Patterns/
```

Offline release artifacts live in `book/releases`.

The generated PDF is published at:

```
https://gturitto.github.io/Agentic-Systems-Patterns/releases/Agentic-Systems-Patterns.pdf
```

License

Single Agent

A single agent receives a goal or message, consults its context, and produces an answer or action. This is the smallest useful unit in the catalog.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

A single agent receives a goal or message, consults its context, and produces an answer or action. This is the smallest useful unit in the catalog.

Use When

- One model-backed worker can complete the task without delegation.
- The interaction has a narrow objective and a clear success condition.
- You want the simplest useful baseline before adding tools, memory, or orchestration.

Avoid When

- The task needs stateful retries, external approvals, multiple specialists, or independent evaluation.
- The agent must recover from long-running failure or resume after interruption.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run single-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`single-agent-pattern/autogen_typescript_example/single_agent.ts`

Open full source

```
// Single Agent Pattern - Autogen TypeScript Example
// To run: npm install && npm run single-agent

import axios from 'axios';
import * as readline from 'readline';
import * as dotenv from 'dotenv';
dotenv.config();

const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';
const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;

async function singleAgent(userInput: string): Promise<string> {
  const response = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny', // or your preferred Mistral model
      messages: [{ role: 'user', content: userInput }],
    },
    {
      headers: {
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
```

```
      },
    }
  );
  return response.data.choices[0].message.content;
}

async function main() {
  const idx = process.argv.indexOf('--input');
  const cliInput = idx !== -1 ? process.argv[idx + 1] : undefined;
  const nonInteractive = cliInput || process.env.NON_INTERACTIVE_INPUT;
  if (nonInteractive) {
    try {
      const agentResponse = await singleAgent(String(nonInteractive));
      console.log('Agent:', agentResponse);
    } catch (err) {
      console.error('Error:', err);
      process.exitCode = 1;
    }
    return;
  }

  const rl = readline.createInterface({ input: process.stdin, output: process.stdout
});
  rl.question('User: ', async (userInput: string) => {
    try {
      const agentResponse = await singleAgent(userInput);
      console.log('Agent:', agentResponse);
    } catch (err) {
      console.error('Error:', err);
    }
    rl.close();
  });
}

main();

// duplicate block removed
```

`single-agent-pattern/langgraph_python_example/single_agent.py`

Open full source

````
# Single Agent Pattern - LangGraph Python Example

This example demonstrates the Single Agent Pattern using LangGraph and Python. The
agent receives a user message, sends it to a Mistral LLM, and returns the response.

## Requirements

- Python 3.8+
- `langgraph` library
- `python-dotenv` (for .env support)
- Mistral LLM API access

## Install dependencies

```bash
pip install langgraph python-dotenv requests
```

## Example Code

```python
import os
from langgraph import Agent, Environment, LLM
from dotenv import load_dotenv

load_dotenv()

MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
MISTRAL_API_URL = "https://api.mistral.ai/v1/chat/completions"

class SimpleEnvironment(Environment):
    def get_observation(self):
        return input("User: ")
    def send_action(self, action):
        print(f"Agent: {action}")

class SingleAgent(Agent):
    def __init__(self, llm):
        self.llm = llm
    def act(self, observation):
        return self.llm.complete(observation)
````
````

```
llm = LLM(
    provider="mistral",
    api_key=MISTRAL_API_KEY,
    api_url=MISTRAL_API_URL,
)

env = SimpleEnvironment()
agent = SingleAgent(llm)

observation = env.get_observation()
action = agent.act(observation)
env.send_action(action)
` ` `


---


- Make sure your `.env` file contains your Mistral API key as `MISTRAL_API_KEY`.
- This is a minimal, functional example.
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `single-agent-pattern/` folder from this repository.

Agent Loop

The agent loop turns a model call into an agent: observe state, decide the next action, act, evaluate the result, and stop when the goal is complete or a limit is reached.

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>
> - Repository source
> - Download code bundle

Intent

The Agent Loop Pattern defines the runtime cycle that turns a model call into an agent: observe state, decide the next action, act through tools or messages, evaluate the result, and stop when the goal is complete or a limit is reached.

Use When

- The next step is not fully known before execution starts.
- The agent may need multiple tool calls or revisions.
- You need explicit stop conditions, budgets, and recoverable state.

Avoid When

- The task is a fixed workflow with known steps.
- A single prompt or deterministic function is sufficient.
- You cannot safely bound tool use, cost, or runtime.

Architecture

```
flowchart LR
  G[Goal] --> O[Observe State]
  O --> D[Decide Next Action]
  D --> A[Act]
  A --> E[Evaluate Result]
  E -->|continue| O
  E -->|done| R[Return Result]
  E -->|budget exceeded| F[Fail Safely]
```

Implementation Notes

- Persist the loop state: goal, messages, observations, tool calls, outputs, errors, and budget counters.
- Make stop conditions explicit: success predicate, max iterations, max cost, max wall-clock time, and user cancellation.
- Treat model output as a proposal. Validate tool names, tool inputs, structured outputs, and permissions before acting.
- Emit trace events for each loop iteration so failures can be debugged after the run.

Failure Modes

- Infinite loops caused by vague goals or missing stop conditions.
- Repeated tool calls with no new information.
- Hidden state drift when observations are summarized too aggressively.
- Premature success when the evaluator only checks whether an answer exists.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `agent-loop-pattern/` folder from this repository.

Related Patterns

- Goals and State
- Tool-Using Agent
- Evaluator-Optimizer

Goals and State

Goals define success; state records progress. Together they make agent work resumable, inspectable, and easier to evaluate.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Goals and State Pattern separates what the agent is trying to achieve from the mutable state it accumulates while working. Goals define success; state records progress, constraints, evidence, and pending work.

Use When

- A task spans multiple turns, tools, or agents.
- You need resumable execution after failure or interruption.
- The agent must explain progress against an explicit objective.

Avoid When

- The task is stateless and can be answered in one call.
- The goal cannot be expressed as observable success criteria.
- State would contain sensitive data you cannot store safely.

Architecture

```
flowchart TD
  U[User Request] --> G[Goal]
  G --> S[Task State]
  S --> P[Planner or Agent Loop]
  P --> W[Work Items]
  W --> S
  S --> C{Goal Complete?}
  C -->|no| P
  C -->|yes| R[Result]
```

Implementation Notes

- Store goals as structured records with `id`, `description`, `success_criteria`, `constraints`, `owner`, and `status`.
- Store state separately from chat history. Chat is evidence; state is the operational model.
- Update state through typed events such as `step_started`, `tool_result`, `blocked`, `approved`, and `completed`.
- Make state transitions auditable and idempotent so a workflow can retry safely.

Failure Modes

- Goals that describe activity rather than success.
- State that becomes a transcript dump instead of a compact task model.
- Agents optimizing for local subgoals that no longer serve the parent goal.
- Lost cancellation or approval state after retries.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `goals-and-state-pattern/` folder from this repository.

Related Patterns

- Agent Loop
- Planning Pattern
- Durable Workflow

Tool Use

Tool use gives an agent controlled access to external capability such as calculators, search, databases, files, code execution, APIs, or business systems.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Tool use gives an agent controlled access to external capability such as calculators, search, databases, files, code execution, APIs, or business systems.

Use When

- The model needs facts, computation, side effects, or system access outside its context window.
- Tool inputs and results can be typed, validated, and observed.
- Permissions and policy checks can sit between model intent and execution.

Avoid When

- A normal function or workflow can solve the problem without model selection.
- The tool has broad destructive permissions and no approval boundary.
- Tool results cannot be verified or traced.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run tool-using-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`tool-using-agent-pattern/autogen_typescript_example/tool_using_agent.ts`

Open full source

```
// Tool-Using Agent Pattern - Autogen TypeScript Example
// To run: npm install && npm run tool-using-agent

import axios from 'axios';
import * as readline from 'readline';
import { evaluate } from 'mathjs';
import * as dotenv from 'dotenv';
dotenv.config();

const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';
const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;

function calculatorTool(input: string): string {
  try {
    const val = evaluate(input);
    return val.toString();
  } catch (e) {
    return `Error: ${e}`;
  }
}

async function toolUsingAgent(userInput: string): Promise<string> {
```

```
  if (userInput.toLowerCase().startsWith('calculate ')) {
    const expr = userInput.substring('calculate '.length);
    return calculatorTool(expr);
  } else {
    const response = await axios.post(
      MISTRAL_API_URL,
      {
        model: 'mistral-tiny',
        messages: [{ role: 'user', content: userInput }],
      },
      {
        headers: {
          'Authorization': `Bearer ${MISTRAL_API_KEY}`,
          'Content-Type': 'application/json',
        },
      }
    );
    return response.data.choices[0].message.content;
  }
}

async function main() {
  const idx = process.argv.indexOf('--input');
  const cliInput = idx !== -1 ? process.argv[idx + 1] : undefined;
  const nonInteractive = cliInput || process.env.NON_INTERACTIVE_INPUT;
  if (nonInteractive) {
    try {
      const agentResponse = await toolUsingAgent(String(nonInteractive));
      console.log('Agent:', agentResponse);
    } catch (err) {
      console.error('Error:', err);
      process.exitCode = 1;
    }
    return;
  }
  const rl = readline.createInterface({ input: process.stdin, output: process.stdout
});
  rl.question('User: ', async (userInput: string) => {
    try {
      const agentResponse = await toolUsingAgent(userInput);
      console.log('Agent:', agentResponse);
```

```
    } catch (err) {
      console.error('Error:', err);
    }
    rl.close();
  });
}

main();
```

`tool-using-agent-pattern/langgraph_python_example/tool_using_agent.py`

Open full source

````
# Tool-Using Agent Pattern - LangGraph Python Example

This example demonstrates the Tool-Using Agent Pattern using LangGraph and Python.
The agent receives a user message, determines if a tool (calculator) is needed, calls
the tool if required, and returns a response. The LLM is Mistral.

## Requirements

- Python 3.8+
- `langgraph` library
- `python-dotenv` (for .env support)
- Mistral LLM API access

## Install dependencies

```bash
pip install langgraph python-dotenv requests
```

## Example Code

```python
import os
from langgraph import Agent, Environment, LLM, Tool
from dotenv import load_dotenv

load_dotenv()

````
````

```python
MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
MISTRAL_API_URL = "https://api.mistral.ai/v1/chat/completions"

import sympy as _sp

def safe_calc(expr: str) -> str:
    try:
        return str(_sp.sympify(expr, evaluate=True))
    except Exception as e:
        return f"Error: {e}"

class CalculatorTool(Tool):
    def call(self, input_str):
        return safe_calc(input_str)

class SimpleEnvironment(Environment):
    def get_observation(self):
        return input("User: ")
    def send_action(self, action):
        print(f"Agent: {action}")

class ToolUsingAgent(Agent):
    def __init__(self, llm, tools):
        self.llm = llm
        self.tools = {tool.name: tool for tool in tools}
    def act(self, observation):
        if observation.lower().startswith("calculate "):
            expr = observation[len("calculate "):]
            return self.tools["calculator"].call(expr)
        else:
            return self.llm.complete(observation)

llm = LLM(
    provider="mistral",
    api_key=MISTRAL_API_KEY,
    api_url=MISTRAL_API_URL,
)

calc_tool = CalculatorTool(name="calculator")
env = SimpleEnvironment()
agent = ToolUsingAgent(llm, [calc_tool])
```

```
observation = env.get_observation()
action = agent.act(observation)
env.send_action(action)
```
```

---

- Type `calculate 2+2` to see the agent use the calculator tool.
- Replace credentials as needed.
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `tool-using-agent-pattern/` folder from this repository.

Structured Output

Structured output constrains model responses to typed data that software can validate and consume.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Structured Output Pattern constrains model responses to typed data that software can validate, route, store, and test. It is the boundary between natural language reasoning and deterministic application logic.

Use When

- Model output controls a tool call, workflow branch, policy decision, or database write.
- Downstream code needs stable fields rather than prose.
- You need regression tests for model-assisted behavior.

Avoid When

- The output is purely creative prose for human reading.
- A deterministic parser already handles the input safely.
- The schema is so broad that it no longer constrains behavior.

Architecture

```
flowchart LR
  P[Prompt] --> M[Model]
  M --> J[JSON or Typed Output]
  J --> V[Schema Validation]
  V -->|valid| D[Deterministic Code]
  V -->|invalid| R[Repair or Refuse]
```

Implementation Notes

- Define schemas close to the code that consumes them.

- Validate every model response before use, even when the provider offers structured output support.
- Prefer enums for routing decisions and discriminated unions for multi-action outputs.
- Log validation failures and repair attempts as first-class evaluation data.

Failure Modes

- Schemas that mirror prose and provide little safety.
- Silent coercion of missing or invalid fields.
- Prompt-only formatting rules with no validator.
- Overly strict schemas that cause brittle failures on harmless variation.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `structured-output-pattern/` folder from this repository.

Related Patterns

- Modern Tool Use
- LLM Router
- Compliance/Policy Enforcer

Context Engineering

Context engineering controls what the model sees: instructions, state, retrieval results, tool documentation, memory, examples, and prior messages.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Context engineering controls what the model sees: instructions, state, retrieval results, tool documentation, memory, examples, and prior messages.

Use When

- Answer quality depends on selecting the right information before generation.
- The agent combines instructions, memory, retrieval, tools, and examples.
- You need to manage context budget, freshness, and source trust.

Avoid When

- All required information is already in a short prompt.
- The system cannot distinguish trusted sources from noisy or stale material.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`context-engineering-pattern/langgraph_python_example/rag_example.py`

Open full source

```
"""
Context Engineering Example: Retrieval-Augmented Generation (RAG) with Mistral
Requirements: pip install langchain-community sentence-transformers faiss-cpu
requests
Note: This is a minimal example using HuggingFace embeddings by default.
"""
import os
import requests
from langchain_community.vectorstores import FAISS
from langchain_community.embeddings import HuggingFaceEmbeddings
from typing import List

# Optionally load environment variables from a local .env if present
try:
    from dotenv import load_dotenv, find_dotenv  # type: ignore
    load_dotenv(find_dotenv(usecwd=True), override=True)
except Exception:
    pass

# Dummy documents
docs = [
    {"content": "Agentic systems are autonomous AI systems."},
    {"content": "Prompt engineering improves LLM outputs."}
]

# Build vector store (in-memory for demo)
texts = [d["content"] for d in docs]
embeddings = HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-
```

```
v2")
try:
    # Try FAISS first (fast vector index)
    vectorstore = FAISS.from_texts(texts, embeddings)
    retriever = vectorstore.as_retriever()

    def retrieve(query: str, k: int = 3):
        return retriever.get_relevant_documents(query)
except Exception:
    # Fallback: simple numpy-based cosine similarity without FAISS
    try:
        import numpy as np  # sentence-transformers depends on numpy, so this should
be available
    except Exception as e:  # pragma: no cover
        raise RuntimeError("numpy is required for the FAISS-free fallback but wasn't
found") from e

    doc_vecs = np.array(embeddings.embed_documents(texts), dtype=float)
    doc_norms = np.linalg.norm(doc_vecs, axis=1, keepdims=True) + 1e-12
    doc_vecs = doc_vecs / doc_norms

    class _MiniDoc:
        def __init__(self, text: str):
            self.page_content = text

    def retrieve(query: str, k: int = 3):
        q = np.array(embeddings.embed_query(query), dtype=float)
        q = q / (np.linalg.norm(q) + 1e-12)
        sims = (doc_vecs @ q)
        top_idx = np.argsort(-sims)[:k]
        return [_MiniDoc(texts[i]) for i in top_idx]
MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
if not MISTRAL_API_KEY:
    raise RuntimeError("Set MISTRAL_API_KEY in your environment.")

def chat_mistral(messages):
    resp = requests.post(
        "https://api.mistral.ai/v1/chat/completions",
        headers={
            "Authorization": f"Bearer {MISTRAL_API_KEY}",
            "Content-Type": "application/json",
```

```
        },
        json={
            "model": "mistral-large-latest",
            "messages": messages,
            "temperature": 0.2,
        },
        timeout=60,
    )
    resp.raise_for_status()
    data = resp.json()
    return (data.get("choices") or [{}])[0].get("message", {}).get("content", "")

if __name__ == "__main__":
    query = "What are agentic systems?"
    retrieved = retrieve(query)
    context = "\n\n".join(d.page_content for d in retrieved)
    answer = chat_mistral([
        {"role": "system", "content": "Use the provided context to answer the
question succinctly."},
        {"role": "user", "content": f"Context:\n{context}\n\nQuestion: {query}"},
    ])
    print("Answer:", answer)
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `context-engineering-pattern/` folder from this repository.

Choosing the Right Pattern

Agentic design is a spectrum. The best architecture is usually the least agentic system that can meet the requirement with acceptable quality, latency, cost, and risk.

Use this chapter before choosing a framework, adding agents, or building a multi-agent topology. The decision should start with the workload, not with the most powerful pattern available.

The Autonomy Ladder

Move up the ladder only when the lower rung cannot handle the job.

Level	Pattern Shape	Use When	Main Risk
1	LLM call	A single answer, rewrite, classification, extraction, or summary is enough.	No access to live data or action.
2	Prompt chain	The work has known phases and each phase can be validated before the next one starts.	Brittle gates or unnecessary latency.
3	Deterministic workflow with LLM steps	Code owns the sequence, while LLMs handle bounded judgment inside steps.	Overfitting the workflow to today's process.
4	Routing and handoffs	Different inputs need different models, prompts, tools, agents, or policies.	Bad routing sends work to the wrong authority.
5	Single agent loop	The next step depends on observations discovered during execution.	Loops, tool misuse, and compounding errors.
6	Orchestrator-workers	The system must break an unknown task into subtasks and synthesize results.	The orchestrator becomes a hidden control plane.
7	Multi-agent system	Different specialists need separate context, tools, permissions, or review roles.	Coordination overhead, trace fragmentation, and cost growth.
8	Autonomous long-running agent	The task spans time, failures, external events, and partial progress.	Unbounded autonomy without strong state, policy, and observability.

This ladder is not a maturity model. A production system can stay at level 2 forever if the task is stable and the value is clear.

First Questions

Ask these questions before selecting a pattern:

- Is the workflow known before execution starts?
- Does the model need to choose the next step, or can code do it?
- Does the task need current data, private data, tools, or side effects?
- Can the system tolerate extra latency from multiple model calls?
- What is the maximum acceptable cost per run?
- What requires human approval?
- What evidence proves the answer or action is correct?
- What state must be replayable after a failure?
- What could go wrong if the model is persuasive but wrong?

If the answers are unclear, start with a deterministic workflow and add agentic behavior only where the workflow needs model judgment.

Selection Matrix

Workload Signal	Prefer	Avoid
Known fixed sequence	Prompt Chaining and Gates	Autonomous agents
One of several known task types	Routing and Handoffs	One giant prompt
Independent subtasks	Parallel Agents	Sequential chains that create avoidable latency
Unknown subtasks	Orchestrator-workers	Hard-coded task lists
Quality improves with review	Evaluator-Optimizer	Single-pass generation
High-risk side effects	Human Approval Gates	Direct tool execution from model output
Large tool surface	MCP-first Tool Use with policy	Broad untyped tools
Long-running work	Durable Workflows and Goals and State	Hidden in-memory loops
Sensitive data or compliance	Policy Enforcement and audit logs	Prompt-only controls
Retrieval-heavy answers	Agentic RAG Systems	Blind model responses
Debugging matters	Observability and Evals	Final-answer-only logs

The same product may combine several rows. For example, a support refund workflow may use routing to classify the request, deterministic workflow steps for account lookup, RAG for policy evidence, human approval for exceptions, and an agent loop only for open-ended investigation.

Workflows vs Agents

A workflow uses code to control the path. A model may classify, extract, summarize, critique, or generate inside a step, but software decides what happens next.

An agent uses the model to decide parts of the path. It observes state, decides the next action, calls tools, reads results, and continues until a goal is complete or a limit is reached.

Prefer workflows when:

- the process is stable;
- correctness depends on deterministic rules;
- latency or cost must stay low;
- the system must be easy to audit;
- operators need predictable failure modes.

Prefer agents when:

- the process is open-ended;
- the system must discover missing information;
- tool choice depends on intermediate observations;
- the number of steps is unknown;
- a fixed workflow would branch into an unmaintainable tree.

Complexity Budget

Every additional agent, model call, tool, memory store, and evaluator spends part of the system's complexity budget. Spend it deliberately.

Add complexity when it buys one of these outcomes:

- higher task completion rate;
- lower human effort;
- better evidence grounding;
- safer side effects;
- lower cost through routing or smaller models;
- better debuggability;
- clearer ownership boundaries.

Do not add complexity only because a pattern is popular. A multi-agent system that replaces a reliable four-step workflow usually makes the product slower, harder to test, and harder to explain.

Pattern Evolution Path

A practical evolution path looks like this:

1. Start with one prompt or deterministic workflow.
2. Add structured outputs and validation.
3. Add retrieval or tools where the model needs evidence or action.
4. Add routing when inputs diverge into distinct paths.
5. Add evaluator loops where quality improves with critique.
6. Add durable state when work spans several steps or sessions.
7. Add agents only where dynamic decisions are required.
8. Add multi-agent coordination when separate roles need separate context, tools, or permissions.

Each step should improve a measured outcome. If a pattern does not improve accuracy, reliability, latency, cost, safety, or maintainability, remove it.

Minimum Production Bar

Before a pattern handles users, money, private data, infrastructure, or customer communication, the system should have:

- typed inputs and outputs;
- explicit stop conditions;
- bounded tool permissions;
- traceable model and tool calls;
- replayable state transitions;
- eval datasets for expected behavior;
- fallback behavior for model, retrieval, and tool failure;
- human approval for high-risk actions;
- rollback or remediation for bad actions.

This minimum bar applies to simple systems too. Small systems fail faster because their boundaries are often implicit.

Related Chapters

- Prompt Chaining and Gates
- Routing and Handoffs

- Circuit Breakers, Fallbacks, and Replay
- Agent Loop
- Goals and State
- Agentic System Architecture
- Source Map

Prompt Chaining and Gates

Prompt chaining breaks a task into known LLM steps and places validation gates between them. It is one of the safest ways to add model judgment without giving the model control over the whole workflow.

Use this pattern when the order of work is known but individual steps need language understanding, generation, classification, or extraction.

Intent

Run several bounded model calls in a fixed sequence. Each call receives a narrow input, produces a typed output, and passes through a deterministic gate before the next step runs.

The model performs local judgment. Code owns the workflow.

Use When

- The task has stable phases.
- Each phase can be tested independently.
- Intermediate outputs can be validated.
- You want lower risk than an autonomous agent loop.
- You need clear failure points and retry behavior.

Common examples:

- classify a request, extract fields, verify fields, then draft a response;
- summarize a document, extract claims, check claims, then produce a final answer;
- generate code, run tests, critique failures, then revise;
- retrieve evidence, normalize citations, then synthesize an answer.

Avoid When

- The steps are unknown before execution starts.
- The chain would branch into dozens of special cases.
- No gate can verify intermediate outputs.
- The same input often needs a different sequence of work.
- Latency from multiple calls is unacceptable.

If the chain keeps growing conditional branches, consider Routing and Handoffs or an Agent Loop.

Architecture

```
Input
  -> Step 1: classify or extract
  -> Gate 1: schema, confidence, policy, completeness
  -> Step 2: transform or retrieve
  -> Gate 2: evidence, consistency, permissions
  -> Step 3: generate final output
  -> Gate 3: final eval, formatting, approval
  -> Result
```

Gates should be deterministic whenever possible. A gate can use a model as an evaluator, but the gate still needs explicit criteria and a clear pass/fail output.

Gate Types

Gate	Checks	Failure Behavior
Schema gate	JSON shape, required fields, enums, numeric ranges	Ask model to repair or fail fast.
Evidence gate	Citations, source freshness, retrieval coverage	Retrieve more evidence or escalate.
Policy gate	permissions, user role, restricted actions	Block, redact, or request approval.
Confidence gate	classification confidence, ambiguity, missing inputs	Ask a clarification question.
Consistency gate	claims match previous state and evidence	Re-run the step with narrower context.
Cost gate	model call count, token budget, time budget	Return partial result or defer.
Human gate	subjective or high-impact decision	Pause until approval.

The strongest chains mix several gate types. A support workflow may use schema gates for extracted fields, policy gates for refunds, and human gates for exceptions.

Implementation Notes

- Keep each prompt narrow. A step should have one job.
- Use structured output for every handoff between steps.
- Persist intermediate outputs, not just the final answer.
- Treat every model output as untrusted until a gate accepts it.
- Make repair loops bounded. A failed schema should not trigger infinite retries.

- Prefer deterministic gates for safety and model-based gates for subjective quality.
- Log each step with input hash, model, output, gate result, latency, and cost.

Example Chain

```
type TicketClass = 'billing' | 'technical' | 'account' | 'unknown';

interface Classification {
  type: TicketClass;
  confidence: number;
  reason: string;
}

interface ExtractedFields {
  customerId?: string;
  orderId?: string;
  issue: string;
}

interface ChainState {
  input: string;
  classification?: Classification;
  fields?: ExtractedFields;
  draft?: string;
}

function gateClassification(result: Classification): string | null {
  if (result.confidence < 0.72) return 'classification_low_confidence';
  if (result.type === 'unknown') return 'unknown_ticket_type';
  return null;
}

function gateFields(fields: ExtractedFields): string | null {
  if (!fields.issue.trim()) return 'missing_issue';
  return null;
}

async function runTicketChain(input: string): Promise<ChainState> {
  const state: ChainState = { input };

  state.classification = await classifyTicket(input);
```

```
  const classificationError = gateClassification(state.classification);
  if (classificationError) throw new Error(classificationError);

  state.fields = await extractTicketFields(input, state.classification.type);
  const fieldError = gateFields(state.fields);
  if (fieldError) throw new Error(fieldError);

  state.draft = await draftResponse(state.classification.type, state.fields);
  return state;
}
```

The example keeps control flow in code. The model classifies, extracts, and drafts, but it does not decide which validation rules apply.

Failure Modes

- A chain pretends to be deterministic while hidden prompts make important decisions.
- Gates validate shape but not meaning.
- The chain passes summarized state forward and loses key evidence.
- Repair loops keep asking the model to fix an output without changing the input or prompt.
- A model-based evaluator accepts fluent but unsupported claims.
- The chain becomes a fragile substitute for a real workflow engine.

Production Checklist

- Are all step inputs and outputs typed?
- Can each step be tested with fixture inputs?
- Does every gate have a named failure reason?
- Does every repair loop have a retry limit?
- Are intermediate outputs persisted for replay?
- Are high-risk actions separated from generation steps?
- Can operators see which gate stopped a run?

Related Chapters

- Choosing the Right Pattern
- Structured Output
- Evaluator-Optimizer
- Human Approval Gates
- Durable Workflows

Routing and Handoffs

Routing sends work to the right model, prompt, tool, workflow, or agent. Handoffs transfer responsibility with enough typed state for the next owner to act safely.

This pattern is often the missing middle between a single prompt and a multi-agent system.

Intent

Use a classifier, policy rule, planner, or deterministic condition to choose the next execution path. Then pass a narrow state object to the selected path.

A router should decide where work belongs. It should not complete the work itself.

Use When

- Inputs belong to distinct domains or intents.
- Different paths need different tools, permissions, models, or prompts.
- Simple requests should use cheap, fast paths.
- Risky requests need stricter policy or human approval.
- A large agent is struggling because it has too many tools or responsibilities.

Common examples:

- route support tickets to billing, technical, account, or fraud workflows;
- route easy requests to a small model and unusual requests to a stronger model;
- route code tasks to search, edit, review, or test agents;
- route requests with side effects through approval workflows;
- route RAG questions to the correct index, tenant, or domain corpus.

Avoid When

- There is only one meaningful path.
- The router cannot explain or score its decision.
- The downstream paths have overlapping ownership.
- A wrong route would create dangerous side effects.
- The router depends on hidden conversation context that is not persisted.

If the route cannot be chosen with confidence, ask for clarification or send the task to a safe fallback path.

Architecture

```
Request
  -> Normalize input
  -> Classify intent, risk, domain, and required capability
  -> Apply policy and budget rules
  -> Select path
  -> Handoff typed state
  -> Downstream path completes or escalates
```

The handoff should include only the state the downstream path needs. Avoid dumping the full conversation into every specialist.

Router Outputs

A router output should be structured:

```
type Route =
  | 'answer_from_docs'
  | 'billing_workflow'
  | 'technical_triage'
  | 'human_review'
  | 'reject'
  | 'clarify';

interface RoutingDecision {
  route: Route;
  confidence: number;
  reason: string;
  requiredCapabilities: string[];
  risk: 'low' | 'medium' | 'high';
}
```

Good router decisions are inspectable. The system should log the route, confidence, reason, model version, policy checks, and fallback behavior.

Handoff Contract

A handoff is not just a message. It is a change in responsibility.

Include:

- normalized user intent;
- route decision and reason;
- relevant extracted fields;
- evidence references;
- permission scope;
- budget remaining;
- deadline or timeout;
- previous failed routes or tool failures;
- expected output contract;
- escalation path.

Exclude:

- unrelated conversation history;
- tools the recipient cannot use;
- hidden policy text that should remain system-owned;
- raw sensitive data when a reference or redacted field is enough.

Router Types

Router Type	Best For	Watch Out For
Rule router	Compliance, tenant, role, known metadata	Rule drift as products change.
Model router	Natural language intent and ambiguous requests	Low confidence and prompt injection.
Embedding router	Domain corpus or knowledge base selection	Similar but unsafe corpora.
Cost router	Model tier selection	Cheap model overuse on hard tasks.
Risk router	Approval, sandbox, or policy paths	Missing high-risk edge cases.
Capability router	Tool or agent selection	Overlapping tool descriptions.

Production systems usually combine them. For example, code may enforce tenant and policy rules before a model chooses between specialist workflows.

Implementation Notes

- Keep route labels stable. Treat route names as API contracts.
- Route before loading large context.
- Separate intent routing from policy routing.
- Use confidence thresholds with explicit clarify or fallback paths.
- Validate handoff state before the recipient starts.

- Do not let downstream agents silently reroute forever.
- Trace route decisions across the whole run.
- Test routers with adversarial and ambiguous examples, not only happy paths.

Failure Modes

- A router becomes a hidden general-purpose agent.
- Route labels overlap, so the same request can fit several paths.
- Handoffs lose authority boundaries and give specialists too many tools.
- Low-confidence routes proceed instead of asking for clarification.
- Multi-agent handoffs create loops with no owner.
- Cost routing sends hard tasks to weak models and hides quality loss.

Production Checklist

- Are route labels mutually understandable and stable?
- Does every route have an owner and output contract?
- Does the router have a fallback for low confidence?
- Does policy run before dangerous handoffs?
- Can an operator explain why a request took a path?
- Are route datasets part of regression tests?
- Are route loops impossible or bounded?

Related Chapters

- Choosing the Right Pattern
- MCP-first Tool Use
- A2A Agent Interoperability
- Supervisor / Worker
- Policy Enforcement
- Context Engineering

Circuit Breakers, Fallbacks, and Replay

Agentic systems need controls that stop waste, contain damage, and make failures explainable. Circuit breakers stop unsafe or unproductive execution. Fallbacks give the system a safer next move. Replay lets engineers reconstruct what happened.

This is a reliability pattern for agent loops, tool use, RAG, workflow orchestration, and multi-agent systems.

Intent

Protect the system from repeated failure and make every run recoverable enough to debug.

An agent should not keep calling the same failing tool, searching the same empty corpus, revising the same bad draft, or handing the same task between agents without progress.

Use When

- Agents can call tools, APIs, browsers, shells, or workflows.
- Work can loop, retry, delegate, or wait.
- Costs can grow with each model or tool call.
- Partial state matters after failure.
- Operators need to explain why a run stopped.

This pattern is mandatory for production systems with side effects.

Avoid When

- The task is a throwaway prototype with no external effect.
- The system has no loop, retry, or tool use.
- Failures do not need investigation.

Even then, keep a simple run log. The prototype that works often becomes the production seed.

Circuit Breakers

A circuit breaker turns repeated or high-risk failure into a stop, fallback, or escalation.

Breaker	Trigger	Action
Tool failure breaker	Same tool fails N times or returns malformed output.	Disable tool for run and choose fallback.

Breaker	Trigger	Action
No-progress breaker	State does not change across iterations.	Stop loop and return blocked status.
Cost breaker	Token, model-call, or money budget is reached.	Stop, summarize progress, and ask for approval.
Latency breaker	Step or run exceeds time budget.	Defer, enqueue, or return partial result.
Retrieval breaker	Search returns low-coverage or conflicting evidence.	Ask clarification or escalate.
Policy breaker	Tool intent violates permissions or risk rules.	Block action and log policy reason.
Handoff breaker	Agents bounce a task between roles.	Assign owner or escalate to human.
Output breaker	Final answer fails schema, citation, or eval checks.	Repair once, then fail safely.

Breakers need names and thresholds. “The agent got stuck” is not operational enough. “No-progress breaker fired after 4 iterations with unchanged evidence set” is debuggable.

Fallbacks

A fallback should be safer than the failed path.

Useful fallback types:

- ask the user for missing information;
- return a partial answer with explicit limits;
- switch from autonomous tool use to deterministic workflow;
- switch from a weak model to a stronger model;
- switch from write action to read-only analysis;
- use cached or previously verified data;
- route to human review;
- schedule background processing;
- fail closed for restricted actions.

Do not use fallback as a way to hide failure. The user or operator should know what changed.

Replay

Replay is the ability to reconstruct a run from durable state.

Store:

- run ID and parent run ID;
- goal and normalized intent;
- model calls, model versions, parameters, and prompt references;
- tool calls, inputs, outputs, errors, and side-effect identifiers;
- retrieval queries, source IDs, and citation metadata;
- state transitions;
- route decisions and handoffs;
- policy checks;
- breaker and fallback events;
- final output and eval results.

Redact sensitive content when required, but keep enough references to investigate the run.

Replay Levels

Level	Capability	Use
Trace replay	Reconstruct what happened from logs.	Incident review and debugging.
Deterministic replay	Re-run deterministic code with recorded model/tool outputs.	Regression tests and workflow validation.
Full replay	Re-run model and tool calls in a sandbox.	Reproduction when external systems are stable.
Counterfactual replay	Re-run with changed prompt, policy, model, or tool.	Evaluate fixes before rollout.

Most teams should start with trace replay. Full replay is useful but harder because models, tools, and external data change.

Implementation Notes

- Add breakers at the loop, tool, retrieval, route, and workflow levels.
- Record breaker events as first-class trace events.
- Tie fallback decisions to explicit failure reasons.
- Separate safe retries from repeated guessing.
- Make every side effect idempotent or traceable.
- Store enough state to resume or fail cleanly.
- Turn incidents into eval cases.

Example Breaker Logic

```
interface ToolFailure {
  tool: string;
```

```
  count: number;
  lastError: string;
}

interface RunBudget {
  maxIterations: number;
  maxToolFailuresPerTool: number;
  maxCostUsd: number;
}

function shouldOpenToolBreaker(
  failure: ToolFailure,
  budget: RunBudget
): boolean {
  return failure.count >= budget.maxToolFailuresPerTool;
}

function shouldStopLoop(iteration: number, costUsd: number, budget: RunBudget):
string | null {
  if (iteration >= budget.maxIterations) return 'max_iterations_reached';
  if (costUsd >= budget.maxCostUsd) return 'max_cost_reached';
  return null;
}
```

The important part is not the code size. The important part is that the stop reason becomes part of the run state and trace.

Failure Modes

- Breakers exist in logs but do not affect execution.
- Fallbacks silently lower quality without telling the user.
- Retries repeat the same inputs and expect different outcomes.
- Traces omit tool inputs, making replay impossible.
- Side effects cannot be matched to the agent run that created them.
- Multi-agent systems have no single owner when a breaker fires.

Production Checklist

- Does every loop have max iterations, max cost, and max wall-clock time?
- Does every tool have timeout, retry, and failure thresholds?
- Does every fallback tell the operator what changed?

- Can the system stop safely after partial progress?
- Can a failed run become an eval fixture?
- Can side effects be audited and reversed when possible?
- Can operators replay a run without reading raw prompts from scattered logs?

Related Chapters

- Agent Loop
- Goals and State
- Self-Healing Workflows
- Durable Workflows
- Observability and Evals
- Policy Enforcement

Source Map

This page maps external references to this book's chapters. Use it as a reading guide, not as a replacement for the book's pattern language.

The sources repeat several core ideas: start simple, separate workflows from agents, use tools through typed contracts, add memory deliberately, evaluate behavior, and reserve multi-agent systems for cases where specialization justifies orchestration cost.

Primary References

Source	Useful Ideas	Book Mapping
Anthropic: Building Effective Agents	Workflows vs agents, prompt chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer, autonomous agents.	Choosing the Right Pattern, Prompt Chaining and Gates, Routing and Handoffs, Evaluator-Optimizer, Agent Loop.
Google Cloud: Choose a design pattern for your agentic AI system	Requirements-first selection, single-agent and multi-agent patterns, sequential, parallel, iterative refinement, human-in-the-loop, custom logic.	Choosing the Right Pattern, Parallel Agents, Human Approval Gates, Agentic System Architecture.
Databricks: Agent system design patterns	Complexity continuum from prompt to deterministic chain to single-agent and multi-agent systems, plus production guidance for testing, tracing, failure handling, model updates, and cost.	Choosing the Right Pattern, Circuit Breakers, Fallbacks, and Replay, Observability and Evals.
MongoDB: 7 Practical Design Patterns for Agentic Systems	Controlled flows, LLM routing, parallelization, reflection, human-in-the-loop, agents, and multi-agent architectures.	Prompt Chaining and Gates, Routing and Handoffs, Reflection, Supervisor / Worker.
Agentic Patterns Graph	Large pattern catalog with categories for context, orchestration, reliability, security, tool use, learning, and UX.	Circuit Breakers, Fallbacks, and Replay, Context Engineering, Working Memory, Policy Enforcement.
GitHub: awesome-agentic-patterns	Curated repository of production and emerging patterns, useful as a discovery index.	This source informs the extended catalog and future additions, especially reliability, context, coding-agent, and tool-use patterns.

Secondary References

Source	Useful Ideas	How To Use
ByteByteGo: Top AI Agentic Workflow Patterns	Accessible explanations of reflection, tool use, ReAct, planning, and multi-agent patterns.	Good introductory reading for readers who want a lighter explanation before the deeper chapters.
Phil Schmid: Zero to One - Learning Agentic Patterns	Practical examples for routing, parallelization, tool use, orchestrator-workers, and multi-agent systems.	Useful for implementation intuition and example shapes.
ProjectPro: Agentic AI Design Patterns	Planning, tool use, reflection, multi-agent examples, and a practical MCP discussion.	Useful for explaining common enterprise examples.
Tungsten Automation: Tool-Use Pattern	Enterprise framing for tool use, ground truth, and workflow APIs.	Maps to MCP-first Tool Use and Policy Enforcement.
Towards AI: 5 Design Patterns in Agentic AI Workflow	Introductory framing for prompt chaining and workflow decomposition.	Useful as a light overview; the article is partially gated.
Medium: Agentic Design Patterns	Common pattern explanations for reflection, tool use, planning, and multi-agent design.	Duplicates core concepts already covered here.
Medium: Agentic Patterns - Architectures for Coordinated AI Systems	Hierarchical, peer-to-peer, market-based, and swarm coordination.	Useful for future expansion of multi-agent coordination patterns.
YouTube: AI agent design patterns	Video walkthrough of agent design patterns.	Use as companion media, not as a primary textual source unless transcript review is added.
YouTube: Master ALL 20 Agentic AI Design Patterns	Broad video catalog of pattern names and examples.	Use as discovery material; validate individual claims against primary sources.

The LinkedIn checkpoint link was not useful as a public source because it requires a login challenge.

Pattern Coverage Map

External Pattern Name	Current Book Chapter
Augmented LLM	Single Agent, Tool Use, Structured Output
Prompt chaining	Prompt Chaining and Gates

External Pattern Name	Current Book Chapter
Routing	Routing and Handoffs
Parallelization	Parallel Agents
Orchestrator-workers	Supervisor / Worker, Planning and Execution
Evaluator-optimizer	Evaluator-Optimizer
Reflection	Reflection
ReAct	ReAct
Agent loop	Agent Loop
Human-in-the-loop	Human Approval Gates
Tool use	Tool Use, MCP-first Tool Use
Memory and context	Context Engineering, Working Memory, Long-Term Episodic Memory
Agentic RAG	Semantic Recall and RAG, Agentic RAG Systems
Multi-agent supervisor	Supervisor / Worker
Peer or network agents	A2A Agent Interoperability, Debate and Consensus
Circuit breaker	Circuit Breakers, Fallbacks, and Replay
Action replay	Circuit Breakers, Fallbacks, and Replay, Observability and Evals
Deterministic chain	Choosing the Right Pattern, Durable Workflows
Coding agent	Coding Agents

Editorial Rule

External sources are used to validate coverage and expose missing patterns. The book keeps its own taxonomy:

- foundations first;
- control loops second;
- memory and knowledge as a separate concern;
- tools, skills, and protocols as integration boundaries;
- multi-agent systems only when specialization has value;
- production runtime patterns for safety, evaluation, and operations;
- source-informed pattern selection as the entry point for choosing the right design.

This keeps the book useful as a reference instead of turning it into a link dump.

Planning and Execution

Planning separates deciding what to do from doing it. The planner creates steps; the executor runs them, reports progress, and handles errors.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Planning separates deciding what to do from doing it. The planner creates steps; the executor runs them, reports progress, and handles errors.

Use When

- The task has meaningful sequencing, dependencies, or recoverable failure points.
- You want to inspect or revise the plan before execution.
- Execution can be deterministic even if planning uses a model.

Avoid When

- The plan would always be a single step.
- The executor cannot report structured progress or failure.
- The model is allowed to execute unvalidated plans directly.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run plan:test
npm run plan:run -- "Compute average of [1,2,3,4]"
npm run plan:py
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`planning-pattern/typescript/src/planner.ts`

Open full source

```
import axios from 'axios';

const MISTRAL_API = 'https://api.mistral.ai/v1/chat/completions';

export interface PlanStep { id: string; description: string }
export interface Plan { steps: PlanStep[]; rationale: string }

export async function planTask(goal: string, apiKey?: string): Promise<Plan> {
  if (!apiKey) {
    // deterministic fallback plan (no network) for tests
    return { steps: [
      { id: 's1', description: 'Load numbers [1,2,3,4]' },
      { id: 's2', description: 'Compute average' }
    ], rationale: 'synthetic' };
  }
  const resp = await axios.post(MISTRAL_API, {
    model: 'mistral-small-latest',
    messages: [
      { role: 'system', content: 'Return JSON {steps: [{id, description}], rationale}
for the goal.' },
```

```
      { role: 'user', content: goal }
    ],
    temperature: 0
  }, { headers: { Authorization: `Bearer ${apiKey}` } });
  const content = resp.data?.choices?.[0]?.message?.content || '';
  try { return JSON.parse(content); } catch { return { steps: [], rationale: content
}; }
}
```

`planning-pattern/typescript/src/executor.ts`

Open full source

```
export async function executePlan(steps: { id: string; description: string }[],
onProgress?: (pct: number, stage: string) => void) {
  const results: Record<string, any> = {};
  for (let i = 0; i < steps.length; i++) {
    const s = steps[i];
    onProgress?.(Math.round((i / steps.length) * 100), s.id);
    // trivial synthetic execution
    if (s.description.includes('Load numbers')) results[s.id] = [1,2,3,4];
    else if (s.description.includes('Compute average')) {
      const arr = results['s1'] || [];
      results[s.id] = arr.reduce((a:number,b:number)=>a+b,0)/arr.length;
    } else results[s.id] = null;
  }
  onProgress?.(100, 'done');
  return results;
}
```

`planning-pattern/typescript/src/run.ts`

Open full source

```
import { planTask } from './planner.js';
import { executePlan } from './executor.js';

async function main() {
  const goal = process.argv.slice(2).join(' ') || 'Compute average of [1,2,3,4]';
  const plan = await planTask(goal, process.env.MISTRAL_API_KEY);
  console.log('Plan:', plan);
```

```
  const results = await executePlan(plan.steps, (pct, stage) =>
console.log('Progress', pct, stage));
  console.log('Results:', results);
}

main();
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `planning-pattern/` folder from this repository.

ReAct

ReAct alternates reasoning and acting. The agent reasons about current state, takes an action, observes the result, and repeats.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

ReAct alternates reasoning and acting. The agent reasons about current state, takes an action, observes the result, and repeats.

Use When

- The task requires tool use and the agent cannot know all required information upfront.
- Observations should change the next step.
- A bounded loop can stop on success, failure, or budget.

Avoid When

- The task is a deterministic workflow with known steps.
- You cannot validate actions before they run.
- The reasoning trace would expose sensitive information to users.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run react-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`react-pattern-reason-act/autogen_typescript_example/react_agent.ts`

Open full source

```
// ReAct Pattern (Reason + Act) - Autogen TypeScript Example
// To run: npm install && npm run react-agent

import axios from 'axios';
import * as readline from 'readline';
import { evaluate } from 'mathjs';
import * as dotenv from 'dotenv';
dotenv.config();

const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';
const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;

function calculatorTool(input: string): string {
  try {
    const val = evaluate(input);
    return val.toString();
  } catch (e) {
    return `Error: ${e}`;
  }
}

async function reactAgent(userInput: string): Promise<string> {
```

```
  let context = userInput;
  let done = false;
  let result = '';
  while (!done) {
    // Step 1: Reason
    const reasoningPrompt = `You are an agent. Think step by step about what to do
next. If you need to use a tool, say TOOL: <expression>. If you are done, say FINAL:
<answer>.\nContext: ${context}`;
    const response = await axios.post(
      MISTRAL_API_URL,
      {
        model: 'mistral-tiny',
        messages: [{ role: 'user', content: reasoningPrompt }],
      },
      {
        headers: {
          'Authorization': `Bearer ${MISTRAL_API_KEY}`,
          'Content-Type': 'application/json',
        },
      }
    );
    const agentOutput = response.data.choices[0].message.content.trim();
    console.log('Agent:', agentOutput);
    if (agentOutput.startsWith('TOOL:')) {
      const expr = agentOutput.replace('TOOL:', '').trim();
      const toolResult = calculatorTool(expr);
      context += `\nTool result: ${toolResult}`;
    } else if (agentOutput.startsWith('FINAL:')) {
      result = agentOutput.replace('FINAL:', '').trim();
      done = true;
    } else {
      // If the agent doesn't follow the protocol, end loop
      result = agentOutput;
      done = true;
    }
  }
  return result;
}

async function main() {
  const idx = process.argv.indexOf('--input');
```

```
  const cliInput = idx !== -1 ? process.argv[idx + 1] : undefined;
  const nonInteractive = cliInput || process.env.NON_INTERACTIVE_INPUT;
  if (nonInteractive) {
    try {
      const agentResponse = await reactAgent(String(nonInteractive));
      console.log('Final Answer:', agentResponse);
    } catch (err) {
      console.error('Error:', err);
      process.exitCode = 1;
    }
    return;
  }
  const rl = readline.createInterface({ input: process.stdin, output: process.stdout
});
  rl.question('Task: ', async (userInput: string) => {
    try {
      const agentResponse = await reactAgent(userInput);
      console.log('Final Answer:', agentResponse);
    } catch (err) {
      console.error('Error:', err);
    }
    rl.close();
  });
}

main();
```

`react-pattern-reason-act/langgraph_python_example/react_agent.py`

Open full source

```
# ReAct Pattern (Reason + Act) - LangGraph Python Example

This example demonstrates the ReAct Pattern using LangGraph and Python. The agent
alternates between reasoning and acting (using a calculator tool), and iteratively
solves the task. The LLM is Mistral.

## Requirements

- Python 3.8+
- `langgraph` library
```

````
- `python-dotenv` (for .env support)
- Mistral LLM API access

## Install dependencies

```bash
pip install langgraph python-dotenv requests
```

## Example Code

```python
import os
from langgraph import Agent, Environment, LLM, Tool
from dotenv import load_dotenv

load_dotenv()

MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
MISTRAL_API_URL = "https://api.mistral.ai/v1/chat/completions"

import sympy as _sp

def safe_calc(expr: str) -> str:
    try:
        return str(_sp.sympify(expr, evaluate=True))
    except Exception as e:
        return f"Error: {e}"

class CalculatorTool(Tool):
    def call(self, input_str):
        return safe_calc(input_str)

class SimpleEnvironment(Environment):
    def get_observation(self):
        return input("Task: ")
    def send_action(self, action):
        print(f"Agent: {action}")

class ReActAgent(Agent):
    def __init__(self, llm, tools):
````
````

````
        self.llm = llm
        self.tools = {tool.name: tool for tool in tools}
    def act(self, observation):
        context = observation
        while True:
            prompt = (
                "You are an agent. Think step by step about what to do next. "
                "If you need to use a tool, say TOOL: <expression>. "
                "If you are done, say FINAL: <answer>.\nContext: " + context
            )
            response = self.llm.complete(prompt)
            print(f"Agent: {response}")
            if response.startswith("TOOL:"):
                expr = response.replace("TOOL:", "").strip()
                tool_result = self.tools["calculator"].call(expr)
                context += f"\nTool result: {tool_result}"
            elif response.startswith("FINAL:"):
                return response.replace("FINAL:", "").strip()
            else:
                return response

llm = LLM(
    provider="mistral",
    api_key=MISTRAL_API_KEY,
    api_url=MISTRAL_API_URL,
)

calc_tool = CalculatorTool(name="calculator")
env = SimpleEnvironment()
agent = ReActAgent(llm, [calc_tool])

observation = env.get_observation()
action = agent.act(observation)
env.send_action(action)
```


---


- Try a multi-step math or logic task to see reasoning and tool use.
- Make sure your `.env` file contains your Mistral API key.
````
````

Download

- Download source bundle
- Open source folder

The download bundle contains the current `react-pattern-reason-act/` folder from this repository.

Reflection

Reflection asks an agent or evaluator to inspect prior output and identify concrete improvements.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Reflection asks an agent or evaluator to inspect prior output and identify concrete improvements.

Use When

- The system has explicit quality criteria.
- A critique can change decisions, tests, state, or final artifacts.
- You need a lightweight improvement pass without a full evaluator-optimizer loop.

Avoid When

- The critique only produces longer output.
- There is no acceptance criterion for the revised result.
- The model is asked to approve its own unsafe actions.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run reflection-self-improvement-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`reflection-and-self-improvement-pattern/autogen_typescript_example/reflection_agent.ts`

Open full source

```
import dotenv from 'dotenv';
dotenv.config();
import axios from 'axios';
import readline from 'readline';

const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;
const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';

if (!MISTRAL_API_KEY) {
  console.error('Please set MISTRAL_API_KEY in your .env file');
  process.exit(1);
}

async function askMistral(messages: any[]) {
  const response = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny',
      messages,
    },
    {
      headers: {
```

```
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    }
  );
  return response.data.choices[0].message.content.trim();
}

async function main() {
  const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout,
  });

  rl.question('Ask the agent a question: ', async (userInput) => {
    let messages = [
      { role: 'system', content: 'You are a helpful assistant that reflects on your answers and tries to improve them if possible.' },
      { role: 'user', content: userInput },
    ];

    // First response
    let answer = await askMistral(messages);
    console.log('\nInitial Answer:\n', answer);

    // Reflection step
    messages.push({ role: 'assistant', content: answer });
    messages.push({ role: 'system', content: 'Reflect on your previous answer. Was it correct, clear, and complete? If not, revise and improve it.' });
    let reflection = await askMistral(messages);
    console.log('\nReflected/Improved Answer:\n', reflection);

    rl.close();
  });
}

main();
```

`reflection-and-self-improvement-pattern/langgraph_python_example/reflection_agent.py`

Open full source

```python
import os
import requests

def ask_mistral(messages):
    api_key = os.getenv('MISTRAL_API_KEY')
    if not api_key:
        raise ValueError('Please set MISTRAL_API_KEY in your .env file')
    url = 'https://api.mistral.ai/v1/chat/completions'
    headers = {
        'Authorization': f'Bearer {api_key}',
        'Content-Type': 'application/json',
    }
    data = {
        'model': 'mistral-tiny',
        'messages': messages,
    }
    response = requests.post(url, headers=headers, json=data)
    response.raise_for_status()
    return response.json()['choices'][0]['message']['content'].strip()

def main():
    user_input = input('Ask the agent a question: ')
    messages = [
        {'role': 'system', 'content': 'You are a helpful assistant that reflects on your answers and tries to improve them if possible.'},
        {'role': 'user', 'content': user_input},
    ]
    # First response
    answer = ask_mistral(messages)
    print('\nInitial Answer:\n', answer)
    # Reflection step
    messages.append({'role': 'assistant', 'content': answer})
    messages.append({'role': 'system', 'content': 'Reflect on your previous answer. Was it correct, clear, and complete? If not, revise and improve it.'})
    reflection = ask_mistral(messages)
    print('\nReflected/Improved Answer:\n', reflection)

if __name__ == '__main__':
    main()
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `reflection-and-self-improvement-pattern/` folder from this repository.

Evaluator-Optimizer

Evaluator-Optimizer pairs a generator with an evaluator. The generator proposes; the evaluator scores; the optimizer revises or stops.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Evaluator-Optimizer Pattern pairs a generator with an evaluator. The generator proposes work; the evaluator scores it against explicit criteria; the optimizer revises until the output passes or the budget ends.

Use When

- Quality can be judged more reliably than it can be produced in one pass.
- You have explicit rubrics, tests, policies, or examples.
- Iterative improvement is worth the extra cost and latency.

Avoid When

- The evaluator is just another vague opinion prompt.
- The task must respond with very low latency.
- You cannot define pass/fail or ranking criteria.

Architecture

```
flowchart LR
  I[Input] --> G[Generator]
  G --> O[Candidate Output]
  O --> E[Evaluator]
  E -->|pass| R[Return]
  E -->|revise with feedback| G
  E -->|budget exhausted| F[Return Best or Fail]
```

Implementation Notes

- Separate generation prompts from evaluation prompts.
- Use deterministic tests when possible, then model-based critique for subjective gaps.
- Persist evaluator feedback so regressions can be analyzed.
- Define max revision count and a fallback behavior before running the loop.

Failure Modes

- Evaluator drift: the evaluator rewards style over correctness.
- Generator overfits to the evaluator and hides flaws.
- Revision loops that make output longer but not better.
- No retained evidence for why a candidate passed.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `evaluator-optimizer-pattern/` folder from this repository.

Related Patterns

- Reflection and Self-Improvement
- Observability and Evals
- Agent Loop

Self-Improvement

Self-improvement uses feedback from prior runs to improve future runs through reviewed changes to prompts, tools, retrieval, policies, tests, or skills.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Self-improvement uses feedback from prior runs to improve future runs through reviewed changes to prompts, tools, retrieval, policies, tests, or skills.

Use When

- You have run logs, eval failures, and review processes.
- Improvements are applied through versioned artifacts.
- Humans or automated gates can approve behavior changes.

Avoid When

- The agent silently rewrites its own instructions in production.
- No eval suite exists to catch regressions.
- The feedback signal is noisy or easy to game.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`reflection-and-self-improvement-pattern/autogen_typescript_example/reflection_agent.ts`

Open full source

```
import dotenv from 'dotenv';
dotenv.config();
import axios from 'axios';
import readline from 'readline';

const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;
const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';

if (!MISTRAL_API_KEY) {
  console.error('Please set MISTRAL_API_KEY in your .env file');
  process.exit(1);
}

async function askMistral(messages: any[]) {
  const response = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny',
      messages,
    },
    {
      headers: {
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    }
  );
```

```
  return response.data.choices[0].message.content.trim();
}

async function main() {
  const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout,
  });

  rl.question('Ask the agent a question: ', async (userInput) => {
    let messages = [
      { role: 'system', content: 'You are a helpful assistant that reflects on your answers and tries to improve them if possible.' },
      { role: 'user', content: userInput },
    ];

    // First response
    let answer = await askMistral(messages);
    console.log('\nInitial Answer:\n', answer);

    // Reflection step
    messages.push({ role: 'assistant', content: answer });
    messages.push({ role: 'system', content: 'Reflect on your previous answer. Was it correct, clear, and complete? If not, revise and improve it.' });
    let reflection = await askMistral(messages);
    console.log('\nReflected/Improved Answer:\n', reflection);

    rl.close();
  });
}

main();
```

`reflection-and-self-improvement-pattern/langgraph_python_example/reflection_agent.py`

Open full source

```
import os
import requests

def ask_mistral(messages):
```

```
    api_key = os.getenv('MISTRAL_API_KEY')
    if not api_key:
        raise ValueError('Please set MISTRAL_API_KEY in your .env file')
    url = 'https://api.mistral.ai/v1/chat/completions'
    headers = {
        'Authorization': f'Bearer {api_key}',
        'Content-Type': 'application/json',
    }
    data = {
        'model': 'mistral-tiny',
        'messages': messages,
    }
    response = requests.post(url, headers=headers, json=data)
    response.raise_for_status()
    return response.json()['choices'][0]['message']['content'].strip()

def main():
    user_input = input('Ask the agent a question: ')
    messages = [
        {'role': 'system', 'content': 'You are a helpful assistant that reflects on your answers and tries to improve them if possible.'},
        {'role': 'user', 'content': user_input},
    ]
    # First response
    answer = ask_mistral(messages)
    print('\nInitial Answer:\n', answer)
    # Reflection step
    messages.append({'role': 'assistant', 'content': answer})
    messages.append({'role': 'system', 'content': 'Reflect on your previous answer. Was it correct, clear, and complete? If not, revise and improve it.'})
    reflection = ask_mistral(messages)
    print('\nReflected/Improved Answer:\n', reflection)

if __name__ == '__main__':
    main()
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `reflection-and-self-improvement-pattern/` folder from this repository.

Self-Healing Workflows

Self-healing workflows detect failed steps and recover through retry, fallback, re-planning, or escalation.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Self-healing workflows detect failed steps and recover through retry, fallback, re-planning, or escalation.

Use When

- Failures are expected and can be classified.
- The workflow can retry idempotently or compensate safely.
- Recovery policies are explicit and observable.

Avoid When

- Every failure requires human judgment.
- Retries can duplicate side effects.
- The system would call a model again without new information.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `self-healing-workflow-agent-pattern/` folder from this repository.

Memory-Augmented Agent

Memory-augmented agents store and retrieve information across turns or sessions.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Memory-augmented agents store and retrieve information across turns or sessions.

Use When

- The agent needs continuity beyond one interaction.
- Stored facts can be scoped, updated, and deleted.
- Retrieval results can be cited or inspected.

Avoid When

- The system would store sensitive data without consent or retention rules.
- Retrieved memories cannot be distinguished from current instructions.
- The memory store is used as an uncurated transcript dump.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run memory-augmented-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`memory-augmented-agent-pattern/autogen_typescript_example/memory_agent.ts`

Open full source

```
import dotenv from 'dotenv';
dotenv.config();
import axios from 'axios';
import readline from 'readline';
import fs from 'fs';

type MemoryMessage = { role: string; content: string };

const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;
const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';
const MEMORY_FILE = './memory-augmented-agent-
pattern/autogen_typescript_example/memory.json';

if (!MISTRAL_API_KEY) {
  console.error('Please set MISTRAL_API_KEY in your .env file');
  process.exit(1);
}

function loadMemory(): MemoryMessage[] {
  if (fs.existsSync(MEMORY_FILE)) {
  return JSON.parse(fs.readFileSync(MEMORY_FILE, 'utf-8')) as MemoryMessage[];
  }
  return [];
}
```

```
function saveMemory(memory: MemoryMessage[]) {
  fs.writeFileSync(MEMORY_FILE, JSON.stringify(memory, null, 2));
}

async function askMistral(messages: any[]) {
  const response = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny',
      messages,
    },
    {
      headers: {
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    }
  );
  return response.data.choices[0].message.content.trim();
}

async function main() {
  const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout,
  });

  let memory: MemoryMessage[] = loadMemory();

  rl.question('Ask the agent a question: ', async (userInput) => {
    // Retrieve relevant memory (for demo, just concatenate all previous
user/assistant messages)
  let context = memory.map((m: MemoryMessage) => `${m.role}:
${m.content}`).join('\n');
    let messages = [
      { role: 'system', content: 'You are a helpful assistant with memory. Use the
following context from previous interactions if relevant:\n' + context },
      { role: 'user', content: userInput },
    ];
```

```
    let answer = await askMistral(messages);
    console.log('\nAgent Answer (with memory):\n', answer);

    // Update memory
  memory.push({ role: 'user', content: userInput });
  memory.push({ role: 'assistant', content: answer });
    saveMemory(memory);

    rl.close();
  });
}

main();
```

`memory-augmented-agent-pattern/langgraph_python_example/memory_agent.py`

Open full source

```
import os
import json
import requests

MEMORY_FILE = os.path.join(os.path.dirname(__file__), 'memory.json')

def load_memory():
    if os.path.exists(MEMORY_FILE):
        with open(MEMORY_FILE, 'r') as f:
            return json.load(f)
    return []

def save_memory(memory):
    with open(MEMORY_FILE, 'w') as f:
        json.dump(memory, f, indent=2)

def ask_mistral(messages):
    api_key = os.getenv('MISTRAL_API_KEY')
    if not api_key:
        raise ValueError('Please set MISTRAL_API_KEY in your .env file')
    url = 'https://api.mistral.ai/v1/chat/completions'
    headers = {
        'Authorization': f'Bearer {api_key}',
```

```
        'Content-Type': 'application/json',
    }
    data = {
        'model': 'mistral-tiny',
        'messages': messages,
    }
    response = requests.post(url, headers=headers, json=data)
    response.raise_for_status()
    return response.json()['choices'][0]['message']['content'].strip()

def main():
    memory = load_memory()
    user_input = input('Ask the agent a question: ')
    # Retrieve relevant memory (for demo, just concatenate all previous
user/assistant messages)
    context = '\n'.join([f"{m['role']}: {m['content']}" for m in memory])
    messages = [
        {'role': 'system', 'content': 'You are a helpful assistant with memory. Use
the following context from previous interactions if relevant:\n' + context},
        {'role': 'user', 'content': user_input},
    ]
    answer = ask_mistral(messages)
    print('\nAgent Answer (with memory):\n', answer)
    # Update memory
    memory.append({'role': 'user', 'content': user_input})
    memory.append({'role': 'assistant', 'content': answer})
    save_memory(memory)

if __name__ == '__main__':
    main()
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `memory-augmented-agent-pattern/` folder from this repository.

Long-Term Episodic Memory

Long-term episodic memory stores events: what happened, when, who was involved, and why it mattered.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Long-term episodic memory stores events: what happened, when, who was involved, and why it mattered.

Use When

- The assistant needs continuity across sessions.
- Events can be retrieved by relevance, recency, user, and project scope.
- You can enforce retention, privacy, and correction policies.

Avoid When

- The task only needs semantic facts, not event history.
- The system cannot explain or delete remembered events.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `long-term-episodic-memory-agent-pattern/` folder from this repository.

Semantic Recall and RAG

Semantic recall retrieves relevant material by meaning rather than exact keywords. RAG injects retrieved material into context before generation.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Semantic recall retrieves relevant material by meaning rather than exact keywords. RAG injects retrieved material into context before generation.

Use When

- The agent must answer from a changing or large knowledge base.
- Relevant documents can be chunked, embedded, filtered, and cited.
- The retrieval layer can enforce source trust and freshness.

Avoid When

- The answer should come from model knowledge only.
- The corpus is too noisy to retrieve safely.
- The system cannot cite or inspect retrieved context.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`context-engineering-pattern/langgraph_python_example/rag_example.py`

Open full source

```
"""
Context Engineering Example: Retrieval-Augmented Generation (RAG) with Mistral
Requirements: pip install langchain-community sentence-transformers faiss-cpu
requests
Note: This is a minimal example using HuggingFace embeddings by default.
"""
import os
import requests
from langchain_community.vectorstores import FAISS
from langchain_community.embeddings import HuggingFaceEmbeddings
from typing import List

# Optionally load environment variables from a local .env if present
try:
    from dotenv import load_dotenv, find_dotenv  # type: ignore
    load_dotenv(find_dotenv(usecwd=True), override=True)
except Exception:
    pass

# Dummy documents
docs = [
    {"content": "Agentic systems are autonomous AI systems."},
    {"content": "Prompt engineering improves LLM outputs."}
]

# Build vector store (in-memory for demo)
texts = [d["content"] for d in docs]
```

```
embeddings = HuggingFaceEmbeddings(model_name="sentence-transformers/all-MiniLM-L6-
v2")
try:
    # Try FAISS first (fast vector index)
    vectorstore = FAISS.from_texts(texts, embeddings)
    retriever = vectorstore.as_retriever()

    def retrieve(query: str, k: int = 3):
        return retriever.get_relevant_documents(query)
except Exception:
    # Fallback: simple numpy-based cosine similarity without FAISS
    try:
        import numpy as np  # sentence-transformers depends on numpy, so this should
be available
    except Exception as e:  # pragma: no cover
        raise RuntimeError("numpy is required for the FAISS-free fallback but wasn't
found") from e

    doc_vecs = np.array(embeddings.embed_documents(texts), dtype=float)
    doc_norms = np.linalg.norm(doc_vecs, axis=1, keepdims=True) + 1e-12
    doc_vecs = doc_vecs / doc_norms

    class _MiniDoc:
        def __init__(self, text: str):
            self.page_content = text

    def retrieve(query: str, k: int = 3):
        q = np.array(embeddings.embed_query(query), dtype=float)
        q = q / (np.linalg.norm(q) + 1e-12)
        sims = (doc_vecs @ q)
        top_idx = np.argsort(-sims)[:k]
        return [_MiniDoc(texts[i]) for i in top_idx]
MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
if not MISTRAL_API_KEY:
    raise RuntimeError("Set MISTRAL_API_KEY in your environment.")

def chat_mistral(messages):
    resp = requests.post(
        "https://api.mistral.ai/v1/chat/completions",
        headers={
            "Authorization": f"Bearer {MISTRAL_API_KEY}",
```

```
            "Content-Type": "application/json",
        },
        json={
            "model": "mistral-large-latest",
            "messages": messages,
            "temperature": 0.2,
        },
        timeout=60,
    )
    resp.raise_for_status()
    data = resp.json()
    return (data.get("choices") or [{}])[0].get("message", {}).get("content", "")


if __name__ == "__main__":
    query = "What are agentic systems?"
    retrieved = retrieve(query)
    context = "\n\n".join(d.page_content for d in retrieved)
    answer = chat_mistral([
        {"role": "system", "content": "Use the provided context to answer the
question succinctly."},
        {"role": "user", "content": f"Context:\n{context}\n\nQuestion: {query}"},
    ])
    print("Answer:", answer)
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `context-engineering-pattern/` folder from this repository.

Working Memory

Working memory is compact, typed task state the agent can update and consult during a run.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Goals and State Pattern separates what the agent is trying to achieve from the mutable state it accumulates while working. Goals define success; state records progress, constraints, evidence, and pending work.

Use When

- A task spans multiple turns, tools, or agents.
- You need resumable execution after failure or interruption.
- The agent must explain progress against an explicit objective.

Avoid When

- The task is stateless and can be answered in one call.
- The goal cannot be expressed as observable success criteria.
- State would contain sensitive data you cannot store safely.

Architecture

```
flowchart TD
  U[User Request] --> G[Goal]
  G --> S[Task State]
  S --> P[Planner or Agent Loop]
  P --> W[Work Items]
  W --> S
  S --> C{Goal Complete?}
  C -->|no| P
  C -->|yes| R[Result]
```

Implementation Notes

- Store goals as structured records with `id`, `description`, `success_criteria`, `constraints`, `owner`, and `status`.
- Store state separately from chat history. Chat is evidence; state is the operational model.
- Update state through typed events such as `step_started`, `tool_result`, `blocked`, `approved`, and `completed`.
- Make state transitions auditable and idempotent so a workflow can retry safely.

Failure Modes

- Goals that describe activity rather than success.
- State that becomes a transcript dump instead of a compact task model.
- Agents optimizing for local subgoals that no longer serve the parent goal.
- Lost cancellation or approval state after retries.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `goals-and-state-pattern/` folder from this repository.

Related Patterns

- Agent Loop
- Planning Pattern
- Durable Workflow

Knowledge-Bound Agents

Knowledge-bound agents ground answers and actions in approved sources, policies, and citation rules.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Knowledge-bound agents ground answers and actions in approved sources, policies, and citation rules.

Use When

- The domain requires approved sources, citations, or compliance constraints.
- The agent should refuse or escalate when evidence is missing.
- Freshness and source trust matter.

Avoid When

- The agent is allowed to speculate freely.
- Approved sources cannot be identified or updated.
- Policy checks happen only after irreversible actions.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `compliance-policy-enforcer-agent/` folder from this repository.

Skills

Skills package procedural knowledge as discoverable folders of instructions, references, scripts, templates, and assets.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Skills Pattern packages procedural knowledge as discoverable folders: concise instructions, references, scripts, templates, and tests that an agent loads only when relevant.

Use When

- A capability requires repeatable domain procedure rather than only a tool API.
- You want reusable know-how across agents, teams, or projects.
- The agent benefits from progressive disclosure: short instructions first, deeper references only when needed.

Avoid When

- A simple tool schema fully describes the capability.
- The skill would embed secrets, credentials, or unsafe scripts.
- The instructions are too vague to test with real tasks.

Architecture

```
flowchart LR
  T[Task] --> D[Skill Discovery]
  D --> S[SKILL.md]
  S --> R[References and Assets]
  S --> C[Scripts or Templates]
  R --> A[Agent Action]
  C --> A
```

Implementation Notes

- Keep `SKILL.md` short and route to deeper files only when needed.
- Bundle scripts and templates instead of asking the model to recreate fragile artifacts.
- Treat skills as supply-chain inputs: review, version, test, and restrict execution.
- Include examples of successful and unsuccessful use.

Failure Modes

- Skill descriptions that are too broad, causing irrelevant activation.
- Long instruction files that consume context before the task is understood.
- Hidden dependencies that only work on one machine.
- Malicious or outdated bundled scripts.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `skills-pattern/` folder from this repository.

Related Patterns

- MCP-first Tool Use
- Context Engineering
- Human-in-the-Loop Approval

MCP-first Tool Use

MCP-first tool use separates tool capability from agent logic through manifests, validation, invocation, and structured results.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Use this pattern when the set of tools may evolve independently from the agent. MCP gives tools a manifest boundary so the agent can inspect capabilities instead of relying on hard-coded assumptions.

Use When

- Tools are shared across agents or applications.
- Tool schemas, context, and permissions need to be discoverable.
- You want to test tool invocation separately from model reasoning.

Avoid When

- A local function call is enough and no discovery boundary is needed.
- Tool inputs cannot be validated.
- The agent has permission to call too many tools without policy checks.

Architecture

```
sequenceDiagram
  participant Agent
  participant MCPServer
  Agent->>MCPServer: Read manifest
  Agent->>Agent: Validate selected tool input
  Agent->>MCPServer: Invoke tool
  MCPServer-->>Agent: Return structured result
```

Implementation Notes

- Validate tool input before every invocation.
- Keep tool results structured and observable.
- Put policy checks between model intent and tool execution.
- Prefer small tools with clear contracts over broad tools with vague descriptions.

Failure Modes

- Tool manifests that describe capabilities too vaguely.
- Model-generated tool inputs used without schema validation.
- Hidden side effects that are not visible in the tool contract.
- No trace linking model decision, tool input, and tool result.

Run the Example

```
npm run mcp:search
npm run mcp:cloud
npm run mcp:agent
npm run mcp:test
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`modern-tool-use-pattern/typescript/src/agent.ts`

Open full source

```
import axios from 'axios';
import Ajv from 'ajv';
import { pathToFileURL } from 'node:url';

const ajv = new Ajv({ allErrors: true, strict: true });

async function getManifest(base: string) {
```

```
  const { data } = await axios.get(`${base}/manifest`);
  return data;
}

async function invoke(base: string, tool: string, input: any) {
  const { data } = await axios.post(`${base}/invoke`, { tool, input });
  if (!data?.ok) throw new Error(data?.error || 'invoke_failed');
  return data.output;
}

export async function runScenario() {
  // Discover tools via MCP manifests
  const search = await getManifest('http://localhost:3031');
  const cloud = await getManifest('http://localhost:3032');
  // Validate inputs before invoking
  const searchSchema = search.tools.find((t: any) => t.name ===
'search.query')?.input_schema;
  const cloudStoreSchema = cloud.tools.find((t: any) => t.name ===
'cloud.store')?.input_schema;
  const vSearch = ajv.compile(searchSchema);
  const vCloudStore = ajv.compile(cloudStoreSchema);
  // Plan (deterministic): search for a topic, then store results in cloud
  const q = 'agents';
  const inputSearch = { q, limit: 2 };
  if (!vSearch(inputSearch)) throw new Error('bad search input');
  const { items } = await invoke('http://localhost:3031', 'search.query',
inputSearch);
  const doc = { topic: q, titles: items.map((i: any) => i.title) };
  const inputStore = { key: `topic:${q}`, value: doc };
  if (!vCloudStore(inputStore)) throw new Error('bad cloud.store input');
  await invoke('http://localhost:3032', 'cloud.store', inputStore);
  return doc;
}

async function main() {
  const out = await runScenario();
  console.log('Stored doc:', out);
}

if (process.argv[1] && import.meta.url === pathToFileURL(process.argv[1]).href) {
```

```
  main().catch(err => { console.error(err); process.exit(1); });
}
```

`modern-tool-use-pattern/typescript/src/mcp_search_server.ts`

Open full source

```
import express from 'express';
import type { Request, Response } from 'express-serve-static-core';

const app = express();
app.use(express.json());

const manifest = {
  name: 'search-tools',
  version: '1.0.0',
  tools: [
    {
      name: 'search.query',
      description: 'Mock web search returning titles for a query',
      input_schema: {
        type: 'object',
        required: ['q'],
        properties: {
          q: { type: 'string' },
          limit: { type: 'number', minimum: 1, maximum: 5 }
        },
        additionalProperties: false
      }
    }
  ]
};

app.get('/manifest', (_req: Request, res: Response) => res.json(manifest));

app.post('/invoke', (req: Request, res: Response) => {
  const { tool, input } = req.body || {};
  if (tool === 'search.query') {
    const q = String(input?.q || '').trim();
    const limit = Math.max(1, Math.min(5, Number(input?.limit ?? 3)));
    const items = Array.from({ length: limit }, (_, i) => ({ title: `${q} result ${i
```

```
+ 1}` }));
    return res.json({ ok: true, output: { items } });
  }
  return res.status(400).json({ ok: false, error: 'unknown_tool' });
});

app.listen(3031, () => console.log('MCP search server on 3031'));
```

`modern-tool-use-pattern/typescript/src/mcp_cloud_server.ts`

Open full source

```
import express from 'express';
import type { Request, Response } from 'express-serve-static-core';

const app = express();
app.use(express.json());

const store = new Map<string, any>();

const manifest = {
  name: 'cloud-store',
  version: '1.0.0',
  tools: [
    {
      name: 'cloud.store',
      description: 'Store a JSON document under a key',
      input_schema: {
        type: 'object',
        required: ['key', 'value'],
        properties: { key: { type: 'string' }, value: { type: 'object' } },
        additionalProperties: false
      }
    },
    {
      name: 'cloud.fetch',
      description: 'Fetch a document by key',
      input_schema: {
        type: 'object',
        required: ['key'],
        properties: { key: { type: 'string' } },
```

```
        additionalProperties: false
      }
    }
  ]
};

app.get('/manifest', (_req: Request, res: Response) => res.json(manifest));

app.post('/invoke', (req: Request, res: Response) => {
  const { tool, input } = req.body || {};
  if (tool === 'cloud.store') {
    store.set(String(input.key), input.value);
    return res.json({ ok: true });
  }
  if (tool === 'cloud.fetch') {
    return res.json({ ok: true, output: { value: store.get(String(input.key)) ?? null
} });
  }
  return res.status(400).json({ ok: false, error: 'unknown_tool' });
});

app.listen(3032, () => console.log('MCP cloud server on 3032'));
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `modern-tool-use-pattern/` folder from this repository.

A2A Agent Interoperability

A2A makes agents discoverable and callable across process, team, runtime, and vendor boundaries.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Use this pattern when one agent needs to call another agent as a remote collaborator. The protocol boundary should make capability discovery, task submission, progress, refusal, cancellation, and results explicit.

Use When

- Agents are owned by different services, teams, runtimes, or vendors.
- A caller must discover what a remote agent can do before sending work.
- Task state must survive asynchronous progress, refusal, error, or cancellation.

Avoid When

- Both agents are simple functions inside one process.
- The interaction is only a local tool call with a typed input and output.
- You cannot authenticate callers or validate messages.

Architecture

```
sequenceDiagram
  participant ClientAgent
  participant AgentCard
  participant RemoteAgent
  ClientAgent->>AgentCard: Fetch capabilities
  ClientAgent->>RemoteAgent: Send task request
  RemoteAgent-->>ClientAgent: Progress, result, refusal, or error
  ClientAgent->>RemoteAgent: Optional cancellation
```

Implementation Notes

- Messages validated against schemas before delivery.
- Bus is in-memory; swap to a real transport without changing messages.
- Treat refusals as valid protocol outcomes, not exceptions.
- Include idempotency keys or task IDs so retries do not duplicate work.
- Add authentication and authorization before crossing a trust boundary.

Failure Modes

- Treating a remote agent like a local tool and ignoring latency, refusal, or cancellation.
- Sending unvalidated natural language blobs instead of typed task messages.
- Missing capability discovery, causing callers to rely on stale assumptions.
- No correlation ID across progress, result, and error messages.

Run the Example

```
npm run a2a:test
npm run a2a:run
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`agent-to-agent-communication-pattern/src/run_demo.ts`

Open full source

```
import { BusMemory } from './bus_memory.ts';
import { AgentA } from './agent_a.ts';
import { AgentB } from './agent_b.ts';

async function run() {
  const bus = new BusMemory();
```

```
  const a = new AgentA(bus);
  const b = new AgentB(bus);
  a.start();
  b.start();
  a.handshake();
  a.requestTask('t1', 'sum', { a: 2, b: 5 });
}

run();
```

agent-to-agent-communication-pattern/src/agent_a.ts

Open full source

```
import { BusMemory, A2A_SCHEMA } from './bus_memory.ts';
import type { Msg } from './bus_memory.ts';
import Ajv from 'ajv';
const ajv = new Ajv({ allErrors: true, strict: true });
const validateResponse = ajv.compile((A2A_SCHEMA as any).properties.TaskResponse);

export class AgentA {
  private bus: BusMemory;
  constructor(bus: BusMemory) { this.bus = bus; }
  start() {
    // listen for responses
    this.bus.subscribe('TaskResponse', (m: Msg) => {
      if (!validateResponse(m.payload)) {
        console.error('Invalid TaskResponse', validateResponse.errors);
        return;
      }
      console.log('AgentA received response:', m.payload);
    });
  }
  handshake() {
    this.bus.publish({ type: 'Handshake', payload: { version: '1.0', capabilities:
['tasks', 'cancel'] } });
  }
  requestTask(id: string, task_type: string, input: any) {
    this.bus.publish({ type: 'TaskRequest', payload: { id, task_type, input, meta: {
ts: Date.now() } } });
  }
```

```
  cancel(id: string, reason: string) {
    this.bus.publish({ type: 'Cancel', payload: { id, reason } });
  }
}
```

`agent-to-agent-communication-pattern/src/agent_b.ts`

Open full source

```
import { BusMemory, A2A_SCHEMA } from './bus_memory.ts';
import type { Msg } from './bus_memory.ts';
import Ajv from 'ajv';
const ajv = new Ajv({ allErrors: true, strict: true });
const validateReq = ajv.compile((A2A_SCHEMA as any).properties.TaskRequest);

export class AgentB {
  private bus: BusMemory;
  private handshakeAckSent = false;
  constructor(bus: BusMemory) { this.bus = bus; }
  start() {
    this.bus.subscribe('Handshake', () => {
      if (this.handshakeAckSent) return;
      this.handshakeAckSent = true;
      this.bus.publish({ type: 'Handshake', payload: { version: '1.0', capabilities:
['tasks'] } });
    });
    this.bus.subscribe('TaskRequest', (m: Msg) => {
      if (!validateReq(m.payload)) return;
      const payload = m.payload as any;
      const { id, task_type, input } = payload;
      if (task_type !== 'sum') {
        this.bus.publish({ type: 'TaskResponse', payload: { id, status: 'refused',
error: 'unsupported_task' } });
        return;
      }
      // progress
      this.bus.publish({ type: 'Progress', payload: { id, stage: 'start', pct: 10,
message: 'starting' } });
      const a = input?.a;
      const b = input?.b;
      if (typeof a !== 'number' || typeof b !== 'number') {
```

```
        this.bus.publish({ type: 'TaskResponse', payload: { id, status: 'error', 
error: 'invalid_input' } });
        return;
      }
      // compute safely
      const sum = a + b;
      this.bus.publish({ type: 'Progress', payload: { id, stage: 'compute', pct: 60 } 
});
      this.bus.publish({ type: 'TaskResponse', payload: { id, status: 'success', 
output: { sum } } });
    });
    this.bus.subscribe('Cancel', (m: Msg) => {
      console.log('AgentB cancel received:', m.payload);
    });
  }
}
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `agent-to-agent-communication-pattern/` folder from this repository.

Secure Agent Communication

Secure communication protects messages between agents with authentication, integrity, confidentiality, and policy checks.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Secure communication protects messages between agents with authentication, integrity, confidentiality, and policy checks.

Use When

- Agents cross trust boundaries.
- Messages contain private data or trigger external side effects.
- You need identity, authorization, replay protection, and audit logs.

Avoid When

- All communication is local and already protected by process boundaries.
- Security checks cannot be enforced before action.
- The protocol lacks correlation IDs and auditability.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run secure-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`secure-agent-communication-pattern/autogen_typescript_example/secure_agent.ts`

Open full source

```
import crypto from 'crypto';
import readline from 'readline';

// Simulated symmetric key for demo (never hardcode in production)
const SECRET_KEY = 'my_demo_secret_key_123';

function encrypt(text: string): string {
  const cipher = crypto.createCipher('aes-256-ctr', SECRET_KEY);
  let encrypted = cipher.update(text, 'utf8', 'hex');
  encrypted += cipher.final('hex');
  return encrypted;
}

function decrypt(encrypted: string): string {
  const decipher = crypto.createDecipher('aes-256-ctr', SECRET_KEY);
  let decrypted = decipher.update(encrypted, 'hex', 'utf8');
  decrypted += decipher.final('utf8');
  return decrypted;
}

async function main() {
  const rl = readline.createInterface({
```

```
    input: process.stdin,
    output: process.stdout,
  });

  rl.question('Agent 1: Enter a message to send securely: ', (message) => {
    const encrypted = encrypt(message);
    console.log('\n[Agent 1] Encrypted message sent:', encrypted);

    // Simulate Agent 2 receiving and decrypting
    const decrypted = decrypt(encrypted);
    console.log('[Agent 2] Decrypted message received:', decrypted);
    rl.close();
  });
}

main();
```

`secure-agent-communication-pattern/langgraph_python_example/secure_agent.py`

Open full source

```
import os
from cryptography.fernet import Fernet

def get_key():
    # For demo, use a static key (never do this in production)
    return Fernet.generate_key()

SECRET_KEY = get_key()
fernet = Fernet(SECRET_KEY)

def encrypt(text):
    return fernet.encrypt(text.encode()).decode()

def decrypt(token):
    return fernet.decrypt(token.encode()).decode()

def main():
    message = input('Agent 1: Enter a message to send securely: ')
    encrypted = encrypt(message)
    print(f'\n[Agent 1] Encrypted message sent: {encrypted}')
```

```python
    # Simulate Agent 2 receiving and decrypting
    decrypted = decrypt(encrypted)
    print(f'[Agent 2] Decrypted message received: {decrypted}')

if __name__ == '__main__':
    main()
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `secure-agent-communication-pattern/` folder from this repository.

Human Approval Gates

Human approval gates pause execution before sensitive, expensive, destructive, or externally visible actions.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Human approval gates pause execution before sensitive, expensive, destructive, or externally visible actions.

Use When

- The agent may trigger irreversible or high-risk actions.
- A human must review context before continuation.
- The workflow can preserve state while waiting.

Avoid When

- Every step requires approval and the agent adds no value.
- Approval records cannot capture who approved what and why.
- The workflow cannot safely resume after waiting.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `human-in-the-loop-approval-agent/` folder from this repository.

Task Delegation

Task delegation assigns bounded subtasks to specialized workers and combines their outputs.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Task delegation assigns bounded subtasks to specialized workers and combines their outputs.

Use When

- Specialized workers can complete independent subtasks better than one general agent.
- The manager can define expected outputs, constraints, and acceptance criteria.
- Subtask results can be merged and checked.

Avoid When

- The task has no useful decomposition.
- Workers share hidden mutable state.
- The manager cannot evaluate returned work.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run task-delegation
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`task-delegation-pattern/autogen_typescript_example/task_delegation.ts`

Open full source

```
// Task Delegation Pattern - Autogen TypeScript Example
// To run: npm install && npm run task-delegation

import axios from 'axios';
import * as readline from 'readline';
import * as dotenv from 'dotenv';
dotenv.config();

const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';
const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;

async function managerAgent(task: string): Promise<string> {
  // Step 1: Decompose the task
  const decomposePrompt = `You are a manager agent. Decompose the following task into two subtasks and describe each one. Task: ${task}`;
  const decomposeResp = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny',
      messages: [{ role: 'user', content: decomposePrompt }],
    },
    {
      headers: {
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
```

```
        'Content-Type': 'application/json',
      },
    }
  );
  const subtasks =
decomposeResp.data.choices[0].message.content.split(/\n|\r/).filter(Boolean);
  const subtask1 = subtasks[0] || 'Subtask 1';
  const subtask2 = subtasks[1] || 'Subtask 2';

  // Step 2: Delegate to workers
  const worker1Prompt = `You are Worker Agent 1. Complete this subtask: ${subtask1}`;
  const worker2Prompt = `You are Worker Agent 2. Complete this subtask: ${subtask2}`;
  const [worker1Resp, worker2Resp] = await Promise.all([
    axios.post(MISTRAL_API_URL, {
      model: 'mistral-tiny',
      messages: [{ role: 'user', content: worker1Prompt }],
    }, {
      headers: {
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    }),
    axios.post(MISTRAL_API_URL, {
      model: 'mistral-tiny',
      messages: [{ role: 'user', content: worker2Prompt }],
    }, {
      headers: {
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    })
  ]);

  // Step 3: Aggregate results
  return `Results:\n- ${worker1Resp.data.choices[0].message.content}\n-
${worker2Resp.data.choices[0].message.content}`;
}

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
```

```
});

rl.question('Task: ', async (userInput: string) => {
  try {
    const result = await managerAgent(userInput);
    console.log(result);
  } catch (err) {
    console.error('Error:', err);
  }
  rl.close();
});
```

`task-delegation-pattern/langgraph_python_example/task_delegation.py`

Open full source

````
# Task Delegation Pattern - LangGraph Python Example

This example demonstrates the Task Delegation Pattern using LangGraph and Python. A
manager agent decomposes a task and delegates subtasks to two worker agents, then
aggregates the results. The LLM is Mistral.

## Requirements

- Python 3.8+
- `langgraph` library
- `python-dotenv` (for .env support)
- Mistral LLM API access

## Install dependencies

```bash
pip install langgraph python-dotenv requests
```

## Example Code

```python
import os
from langgraph import Agent, Environment, LLM
from dotenv import load_dotenv
````
````

```
load_dotenv()

MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
MISTRAL_API_URL = "https://api.mistral.ai/v1/chat/completions"

class SimpleEnvironment(Environment):
    def get_observation(self):
        return input("Task: ")
    def send_action(self, action):
        print(action)

class ManagerAgent(Agent):
    def __init__(self, llm):
        self.llm = llm
    def act(self, task):
        prompt = f"You are a manager agent. Decompose the following task into two
subtasks and describe each one. Task: {task}"
        return self.llm.complete(prompt)

class WorkerAgent(Agent):
    def __init__(self, llm, name):
        self.llm = llm
        self.name = name
    def act(self, subtask):
        prompt = f"You are {self.name}. Complete this subtask: {subtask}"
        return self.llm.complete(prompt)

llm = LLM(
    provider="mistral",
    api_key=MISTRAL_API_KEY,
    api_url=MISTRAL_API_URL,
)

env = SimpleEnvironment()
manager = ManagerAgent(llm)
worker1 = WorkerAgent(llm, "Worker Agent 1")
worker2 = WorkerAgent(llm, "Worker Agent 2")

task = env.get_observation()
subtasks = manager.act(task).split("\n")
```

```
subtask1 = subtasks[0] if len(subtasks) > 0 else "Subtask 1"
subtask2 = subtasks[1] if len(subtasks) > 1 else "Subtask 2"
result1 = worker1.act(subtask1)
result2 = worker2.act(subtask2)
final = f"Results:\n- {result1}\n- {result2}"
env.send_action(final)
```

- Try a complex or multi-step task to see delegation in action.
- Make sure your `.env` file contains your Mistral API key.
```

## Download

- Download source bundle
- Open source folder

The download bundle contains the current `task-delegation-pattern/` folder from this repository.
```

Supervisor / Worker

Supervisor/Worker centralizes goal ownership, task state, routing, and quality gates while workers perform bounded specialist work.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Supervisor/Worker centralizes goal ownership, task state, routing, and quality gates while workers perform bounded specialist work.

Use When

- Independent specialists are useful but centralized control is still required.
- The supervisor can route work and evaluate outputs.
- Workers have narrow roles and structured return contracts.

Avoid When

- The supervisor becomes a bottleneck with no added quality control.
- Workers can take uncontrolled side effects.
- No one owns final acceptance.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run hierarchical-agent
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`hierarchical-agent-pattern/autogen_typescript_example/hierarchical_agent.ts`

Open full source

```
import dotenv from 'dotenv';
dotenv.config();
import axios from 'axios';
import readline from 'readline';

const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;
const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';

if (!MISTRAL_API_KEY) {
  console.error('Please set MISTRAL_API_KEY in your .env file');
  process.exit(1);
}

async function askMistral(messages: any[]) {
  const response = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny',
      messages,
    },
    {
      headers: {
```

```
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    }
  );
  return response.data.choices[0].message.content.trim();
}

async function main() {
  const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout,
  });

  rl.question('State the overall goal for the manager agent: ', async (goalInput) =>
{
    // Manager agent decomposes the goal into two sub-tasks
    let managerMessages = [
      { role: 'system', content: 'You are a manager agent. Decompose the user goal
into two sub-tasks and assign each to a worker agent.' },
      { role: 'user', content: `Goal: ${goalInput}` },
    ];
    let managerPlan = await askMistral(managerMessages);
    console.log('\nManager Agent Plan:\n', managerPlan);

    // Simulate two worker agents (for demo, just ask Mistral for each sub-task)
    const subTasks = managerPlan.match(/Sub-task [12]: (.*)/g) || [];
    let workerResults: string[] = [];
    for (let i = 0; i < subTasks.length; i++) {
      let workerMessages = [
        { role: 'system', content: `You are Worker Agent ${i+1}. Complete the
following sub-task as best as you can.` },
        { role: 'user', content: subTasks[i] },
      ];
      let workerResult = await askMistral(workerMessages);
      console.log(`\nWorker Agent ${i+1} Result:\n`, workerResult);
      workerResults.push(workerResult);
    }

    // Manager aggregates results
    let aggregationMessages = [
```

```
      { role: 'system', content: 'You are a manager agent. Aggregate the following
worker results into a final answer for the user.' },
      { role: 'user', content: workerResults.map((r, i) => `Worker ${i+1}:
${r}`).join('\n') },
    ];
    let finalResult = await askMistral(aggregationMessages);
    console.log('\nFinal Aggregated Result for User:\n', finalResult);

    rl.close();
  });
}

main();
```

`hierarchical-agent-pattern/langgraph_python_example/hierarchical_agent.py`

Open full source

```
import os
import requests
import re

def ask_mistral(messages):
    api_key = os.getenv('MISTRAL_API_KEY')
    if not api_key:
        raise ValueError('Please set MISTRAL_API_KEY in your .env file')
    url = 'https://api.mistral.ai/v1/chat/completions'
    headers = {
        'Authorization': f'Bearer {api_key}',
        'Content-Type': 'application/json',
    }
    data = {
        'model': 'mistral-tiny',
        'messages': messages,
    }
    response = requests.post(url, headers=headers, json=data)
    response.raise_for_status()
    return response.json()['choices'][0]['message']['content'].strip()

def main():
    goal_input = input('State the overall goal for the manager agent: ')
```

```
    # Manager agent decomposes the goal into two sub-tasks
    manager_messages = [
        {'role': 'system', 'content': 'You are a manager agent. Decompose the user
goal into two sub-tasks and assign each to a worker agent.'},
        {'role': 'user', 'content': f'Goal: {goal_input}'},
    ]
    manager_plan = ask_mistral(manager_messages)
    print('\nManager Agent Plan:\n', manager_plan)

    # Simulate two worker agents (for demo, just ask Mistral for each sub-task)
    sub_tasks = re.findall(r'Sub-task [12]: (.*)', manager_plan)
    worker_results = []
    for i, sub_task in enumerate(sub_tasks):
        worker_messages = [
            {'role': 'system', 'content': f'You are Worker Agent {i+1}. Complete the
following sub-task as best as you can.'},
            {'role': 'user', 'content': sub_task},
        ]
        worker_result = ask_mistral(worker_messages)
        print(f'\nWorker Agent {i+1} Result:\n', worker_result)
        worker_results.append(worker_result)

    # Manager aggregates results
    aggregation_messages = [
        {'role': 'system', 'content': 'You are a manager agent. Aggregate the
following worker results into a final answer for the user.'},
        {'role': 'user', 'content': '\n'.join([f'Worker {i+1}: {r}' for i, r in
enumerate(worker_results)])},
    ]
    final_result = ask_mistral(aggregation_messages)
    print('\nFinal Aggregated Result for User:\n', final_result)

if __name__ == '__main__':
    main()
```

Download

- Download source bundle
- Open source folder

The download bundle contains the current `hierarchical-agent-pattern/` folder from this repository.

Debate and Consensus

Debate and consensus use multiple independent proposals, critiques, votes, or rankings before producing a final answer.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Debate and consensus use multiple independent proposals, critiques, votes, or rankings before producing a final answer.

Use When

- Diversity of reasoning improves quality.
- Aggregation rules are clear before agents start.
- The system can tolerate extra cost and latency.

Avoid When

- Agents are not independent and will repeat the same failure.
- Voting replaces evidence or tests.
- The final decision has no accountable owner.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `consensus-seeking-multi-agent-system-pattern/` folder from this repository.

Parallel Agents

Parallel agents run independent work concurrently, then merge results through a fan-out/fan-in control point.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Parallel agents run independent work concurrently, then merge results through a fan-out/fan-in control point.

Use When

- Work can be split into independent searches, reviews, or candidate generations.
- Latency matters and parallelism is safe.
- The merge step can compare, rank, or synthesize results.

Avoid When

- Agents need shared mutable state during execution.
- The merge policy is vague.
- Parallel work increases cost without increasing quality.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Run the Example

```
npm run multi-agent-collab
```

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

These excerpts show the implementation shape. The complete code is available in the download bundle and repository source.

`multi-agent-collaboration-pattern/autogen_typescript_example/multi_agent_collab.ts`

Open full source

```
// Multi-Agent Collaboration Pattern - Autogen TypeScript Example
// To run: npm install && npm run multi-agent-collab

import axios from 'axios';
import * as readline from 'readline';
import * as dotenv from 'dotenv';
dotenv.config();

const MISTRAL_API_URL = 'https://api.mistral.ai/v1/chat/completions';
const MISTRAL_API_KEY = process.env.MISTRAL_API_KEY;

async function agent(name: string, role: string, input: string): Promise<string> {
  const prompt = `You are ${name}, your role is: ${role}. Here is your input:
${input}`;
  const response = await axios.post(
    MISTRAL_API_URL,
    {
      model: 'mistral-tiny',
      messages: [{ role: 'user', content: prompt }],
    },
    {
      headers: {
```

```
        'Authorization': `Bearer ${MISTRAL_API_KEY}`,
        'Content-Type': 'application/json',
      },
    }
  );
  return response.data.choices[0].message.content;
}

async function multiAgentCollab(task: string) {
  // Agent 1: Idea Generator
  const idea = await agent('Alice', 'Idea Generator', task);
  console.log('Alice (Idea Generator):', idea);

  // Agent 2: Critic
  const critique = await agent('Bob', 'Critic', `Here is an idea: ${idea}\nPlease
critique or improve it.`);
  console.log('Bob (Critic):', critique);

  // Agent 1: Finalize
  const final = await agent('Alice', 'Idea Generator', `Here is the critique:
${critique}\nPlease finalize the solution.`);
  console.log('Alice (Finalized):', final);
}

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

rl.question('Task: ', async (userInput: string) => {
  try {
    await multiAgentCollab(userInput);
  } catch (err) {
    console.error('Error:', err);
  }
  rl.close();
});
```

`multi-agent-collaboration-pattern/langgraph_python_example/multi_agent_collab.py`

Open full source

````markdown
# Multi-Agent Collaboration Pattern - LangGraph Python Example

This example demonstrates the Multi-Agent Collaboration Pattern using LangGraph and Python. Two agents (an Idea Generator and a Critic) collaborate to solve a task, exchanging messages and refining the solution. The LLM is Mistral.

## Requirements

- Python 3.8+
- `langgraph` library
- `python-dotenv` (for .env support)
- Mistral LLM API access

## Install dependencies

```bash
pip install langgraph python-dotenv requests
```

## Example Code

```python
import os
from langgraph import Agent, Environment, LLM
from dotenv import load_dotenv

load_dotenv()

MISTRAL_API_KEY = os.getenv("MISTRAL_API_KEY")
MISTRAL_API_URL = "https://api.mistral.ai/v1/chat/completions"

class SimpleEnvironment(Environment):
    def get_observation(self):
        return input("Task: ")
    def send_action(self, action):
        print(action)

class IdeaGenerator(Agent):
    def __init__(self, llm):
        self.llm = llm
    def act(self, observation):
````
````

````
        prompt = f"You are Alice, an Idea Generator. Here is your task: {observation}"
        return self.llm.complete(prompt)

class Critic(Agent):
    def __init__(self, llm):
        self.llm = llm
    def act(self, idea):
        prompt = f"You are Bob, a Critic. Here is an idea: {idea}\nPlease critique or improve it."
        return self.llm.complete(prompt)

llm = LLM(
    provider="mistral",
    api_key=MISTRAL_API_KEY,
    api_url=MISTRAL_API_URL,
)

env = SimpleEnvironment()
idea_agent = IdeaGenerator(llm)
critic_agent = Critic(llm)

task = env.get_observation()
idea = idea_agent.act(task)
print("Alice (Idea Generator):", idea)
critique = critic_agent.act(idea)
print("Bob (Critic):", critique)
final = idea_agent.act(f"Here is the critique: {critique}\nPlease finalize the solution.")
print("Alice (Finalized):", final)
```


---


- Try a creative or open-ended task to see agent collaboration.
- Make sure your `.env` file contains your Mistral API key.
````

Download

- Download source bundle
- Open source folder
````

The download bundle contains the current `multi-agent-collaboration-pattern/` folder from this repository.

CrewAI Flows and Crews

CrewAI Flows own state and execution order. Crews group specialized agents that collaborate on delegated work inside the flow.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The CrewAI Flows and Crews Pattern separates production workflow control from collaborative agent work. Flows own state and execution order; crews group specialized agents that perform delegated tasks inside the flow.

Use When

- You are building Python-first agent automations.
- The system needs explicit state and event-driven control flow.
- Multiple specialist agents need to collaborate on bounded tasks.

Avoid When

- A single deterministic workflow step is enough.
- Agents have unclear roles or overlapping responsibilities.
- You cannot define where flow state begins and crew-local context ends.

Architecture

```
flowchart TD
  Event[Event or User Request] --> Flow[CrewAI Flow]
  Flow --> State[Flow State]
  Flow --> Crew[Crew]
  Crew --> A1[Researcher Agent]
  Crew --> A2[Writer Agent]
  Crew --> A3[Reviewer Agent]
  Crew --> Result[Crew Result]
  Result --> State
  State --> Output[Final Output]
```

Implementation Notes

- Let flows manage state, branching, persistence, and execution order.
- Give each crew a bounded task with clear expected output.
- Give each agent a role that changes behavior, not just a different name.
- Test flow state transitions separately from crew output quality.

Failure Modes

- Crews used as a substitute for workflow design.
- Too many agents with vague roles.
- Flow state mutated implicitly through chat history.
- No evaluator for whether the crew result satisfies the flow step.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `crewai-flows-and-crews-pattern/` folder from this repository.

Related Patterns

- Task Delegation
- Consensus-Seeking Multi-Agent System
- Durable Workflow

Agentic System Architecture

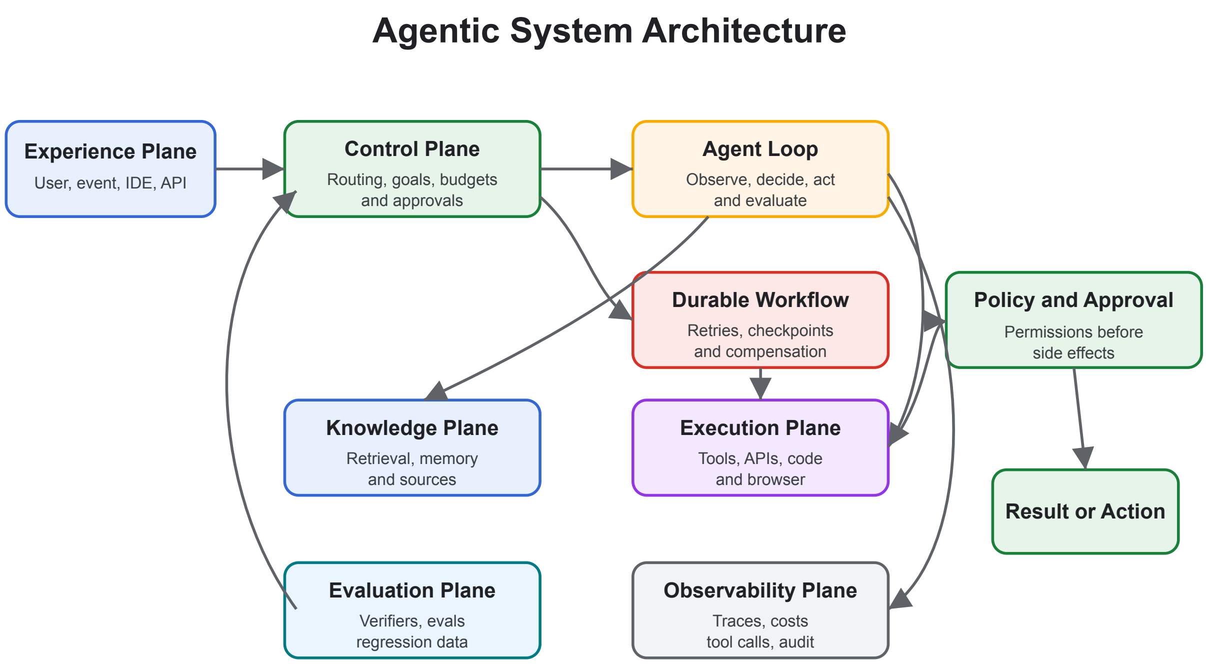
An agentic system is more than a model call wrapped in a loop. It is a set of control planes, execution planes, data planes, and safety planes that let software use model judgment without losing engineering control.

Use this chapter when a single pattern is not enough and you need to combine agents, tools, memory, policies, workflows, evals, and observability into one coherent system.

Core Idea

Separate the system into planes:

- **Experience plane:** chat, IDE, API, webhook, ticket, mobile, or scheduled entrypoint.
- **Control plane:** routing, planning, permissions, budgets, approvals, task state, and stop conditions.
- **Execution plane:** tools, code execution, browser actions, API calls, workflows, and external side effects.
- **Knowledge plane:** retrieval, memory, indexes, metadata, source freshness, and citations.
- **Evaluation plane:** offline evals, runtime checks, verifiers, red-team tests, and regression datasets.
- **Observability plane:** traces, costs, latency, tool calls, model inputs, decisions, and operator review.



Architecture Questions

- What owns the goal?
- What owns state?
- What may call tools?
- What requires approval?
- What happens when evidence is missing?
- What happens when the model is wrong?
- What can be replayed after a failure?
- What is deterministic, and what is model-mediated?

The system should have direct answers to those questions before it handles private data, money movement, production infrastructure, or customer-facing communication.

Boundary Design

The most important architectural choice is the boundary between model judgment and deterministic software. Keep model outputs as proposals until software validates them.

Good boundaries include:

- Typed tool schemas
- Policy checks before side effects
- Explicit state transitions

- Human approval for high-risk operations
- Retrieval filters and citations
- Budget and timeout limits
- Audit logs that connect prompt, decision, tool input, and result

Weak boundaries include:

- Broad shell or browser access without approval
- Prompt-only policy enforcement
- Unstructured memory writes
- Hidden retries
- Tool results that cannot be traced
- Agents that can rewrite their own operating rules without review

Composition Patterns

Common production systems combine several chapters from this book:

- **Agent Loop** for bounded observe-decide-act cycles.
- **Goals and State** for resumability and auditability.
- **MCP-first Tool Use** for discoverable tools.
- **Agentic RAG Systems** for evidence-grounded answers.
- **Durable Workflows** for long-running state, retries, and approvals.
- **Observability and Evals** for quality gates and regression control.
- **Policy Enforcement** for permission and compliance checks.

Failure Modes

- The model becomes the control plane and no deterministic component owns state or permissions.
- Every pattern is added at once, producing a system that is powerful but impossible to debug.
- Retrieval, memory, and tool output are mixed into one untrusted context blob.
- The system has no replay path after a bad action.
- Evals are added after production failures instead of before launch.

Design Rule

Architecture should make failure visible. If a run fails, an operator should be able to answer: what goal was active, what evidence was used, what tool calls ran, what policy checks passed, what changed, and why the system stopped.

Related Chapters

- Agent Loop
- Goals and State
- MCP-first Tool Use
- Agentic RAG Systems
- Durable Workflows
- Observability and Evals

Agentic RAG Systems

Agentic RAG uses agent behavior around retrieval. Instead of one retrieve-then-generate call, the system can plan queries, choose sources, call retrieval tools, inspect evidence, refine searches, verify citations, and decide whether to answer, ask for clarification, or escalate.

This chapter is architectural. The existing Semantic Recall and RAG chapter explains the retrieval pattern. This chapter explains how to build a full system around it.

Basic RAG vs Agentic RAG

Basic RAG is a pipeline:

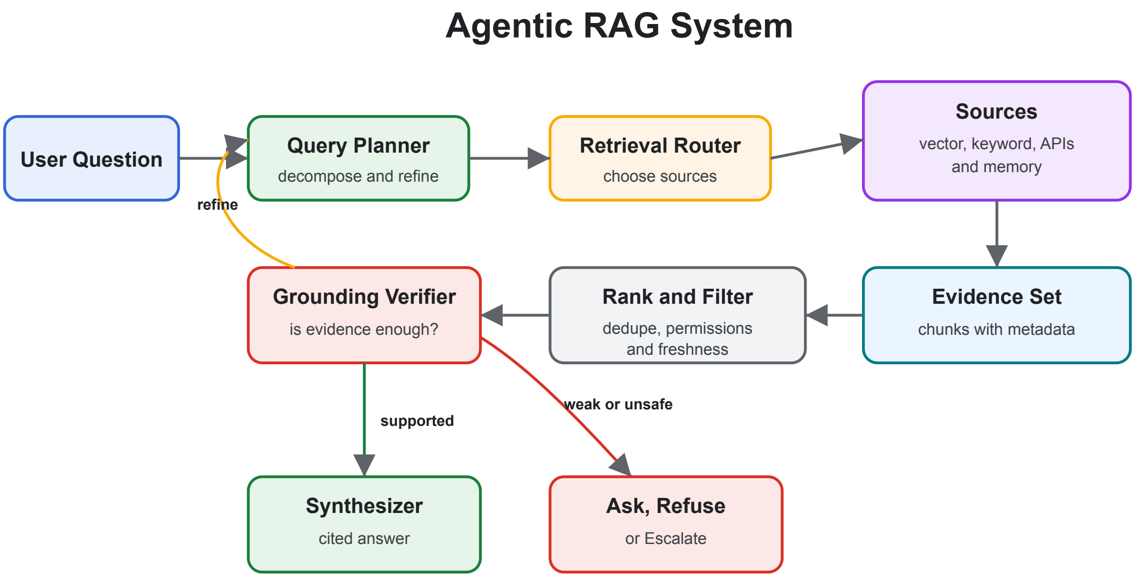
```
query -> retrieve -> inject context -> generate answer
```

Agentic RAG is a control loop:

```
goal -> plan retrieval -> search -> inspect evidence -> refine or verify -> answer or refuse
```

The shift matters because many real questions are not one search. They require decomposition, source selection, permissions, freshness checks, and answer verification.

Reference Flow



System Components

- **Query planner:** Decomposes complex requests into retrieval tasks.
- **Retrieval router:** Chooses indexes, databases, APIs, memories, or web sources.
- **Access filter:** Applies user permissions and source-level policy before retrieval.
- **Retriever:** Runs vector, keyword, graph, SQL, API, or hybrid search.
- **Ranker:** Removes duplicates, stale chunks, low-relevance evidence, and policy-ineligible content.
- **Verifier:** Checks whether the evidence actually supports the answer.
- **Synthesizer:** Produces the final answer with citations and uncertainty.
- **Telemetry:** Records query, sources, ranks, citations, costs, and verifier outcomes.

Use When

- The answer must be grounded in changing or private sources.
- The question may require multiple searches or source types.
- Users need citations, provenance, and freshness.
- Retrieval failures should lead to clarification, refusal, or escalation instead of hallucination.

Avoid When

- A deterministic database query would answer the question directly.
- The system cannot enforce source permissions.

- The corpus is too noisy to support reliable grounding.
- Citation quality is not evaluated.

Architecture Decisions

Define these decisions explicitly:

- Source inventory: which sources exist and who owns them.
- Freshness policy: how stale each source may be.
- Chunking policy: how documents are split and updated.
- Metadata policy: tenant, user, project, date, sensitivity, and source type.
- Retrieval strategy: vector, keyword, graph, SQL, API, or hybrid.
- Citation policy: what counts as a valid citation.
- Refusal policy: what happens when evidence is weak.
- Evaluation policy: what dataset proves retrieval quality.

Corrective RAG Loop

Corrective RAG adds a verifier before final synthesis. If the evidence is weak, the system changes the query or source rather than forcing an answer.

The diagram above shows this loop: weak evidence returns to planning, supported evidence moves to synthesis, and unsafe or missing evidence becomes a refusal, clarification, or escalation path.

Multi-Agent RAG

A multi-agent RAG system separates roles:

- Researcher: decomposes the information need.
- Retriever: searches specific sources.
- Verifier: checks evidence against claims.
- Synthesizer: writes the answer.
- Policy agent: checks source eligibility and sensitive-data boundaries.

Use separate agents only when the roles produce independent value. If all roles read the same evidence and repeat the same prompt, use one agent with structured steps.

Failure Modes

- Hallucinated citations: the answer cites a source that does not support the claim.
- Retrieval drift: the loop keeps searching adjacent topics instead of the user question.
- Over-fetching: too much context pushes out the important evidence.

- Permission leaks: retrieval returns documents the user should not see.
- Stale grounding: old documentation wins because it is easier to retrieve.
- Memory contamination: user preferences are treated as factual source material.

Production Controls

- Access-controlled indexes
- Metadata filters before retrieval
- Evidence-count and freshness thresholds
- Citation validation
- Retrieval eval datasets
- Source-level telemetry
- Human escalation for conflicting evidence
- Red-team tests for prompt injection in retrieved documents

Related Chapters

- Semantic Recall and RAG
- Context Engineering
- Knowledge-Bound Agents
- Evaluator-Optimizer
- Observability and Evals

Open Personal Agent Architectures

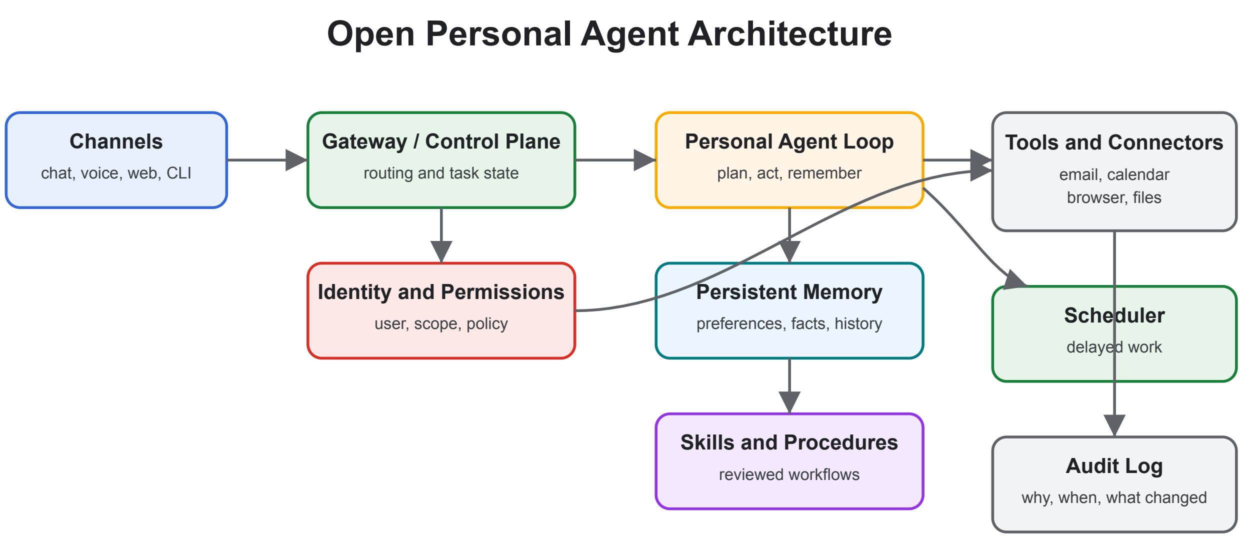
Open personal agents are self-hosted or user-controlled assistants that connect to chat apps, calendars, inboxes, browsers, files, memory, and automations. OpenClaw and Hermes Agent are useful examples because they emphasize different sides of the architecture: broad personal action versus long-term learning.

This chapter is not a product ranking. It uses these projects to explain the architecture of persistent personal agents.

Examples

- OpenClaw and its GitHub repository: a personal assistant that runs on user-controlled infrastructure and acts through channels such as chat apps, device surfaces, and connected tools.
- Hermes Agent and its GitHub repository: a persistent agent focused on memory, skill creation, and learning from repeated use.
- AutoGPT: a platform for creating and running continuous agents.
- OpenHands: an open-source platform for software-development agents.

Core Architecture



What Makes Them Different

A normal assistant answers in the current session. A personal agent keeps context and can act over time. That creates value and risk.

Useful capabilities:

- Persistent user preferences
- Project and relationship memory
- Scheduled work
- App integrations
- Reusable skills
- Multi-channel access
- Long-running tasks

New risks:

- Over-broad OAuth scopes
- Mistaken identity in email or chat
- Memory storing private or incorrect facts
- Automation without enough review
- Prompt injection through inboxes, calendars, pages, and documents
- Operational burden when self-hosted

OpenClaw-Style Pattern

OpenClaw-style systems optimize for reach: one assistant connected to many channels and tools.

Architecture emphasis:

- Gateway-first design
- Chat-native interaction
- Tool and app connectors
- User-owned deployment
- Broad task automation

Use this style when the main problem is giving a user one assistant across daily applications.

Hermes-Style Pattern

Hermes-style systems optimize for learning over time.

Architecture emphasis:

- Persistent memory
- Skill extraction from repeated workflows
- Self-improvement loops
- Long-running agent presence
- User model that deepens over sessions

Use this style when the main problem is making the assistant better at one user's recurring workflows.

Safety Architecture

Personal agents need stronger safety boundaries than chat-only assistants because they can touch private systems.

Minimum controls:

- Identity verification for requests from email, chat, and shared channels
- Tool scopes narrowed by task type
- Human approval before sending messages, spending money, changing access, or deleting data
- Separate memory types for preferences, facts, credentials, and task state
- Audit logs for every external action
- Prompt-injection filters for retrieved documents, emails, and webpages
- Secrets stored outside model-visible context

Failure Modes

- The agent trusts a spoofed sender and acts on private data.
- A chat instruction overrides the user's standing policy.
- The agent stores a temporary fact as durable memory.
- A connector exposes more data than the task needs.
- The user cannot inspect why an action happened.
- The system has no emergency stop or approval mode.

Design Rule

Treat the personal agent like an employee with keys. It needs identity, permissions, training, supervision, logs, and a way to revoke access.

Related Chapters

- Skills

- Memory-Augmented Agent
- Human Approval Gates
- Policy Enforcement
- Agentic System Architecture

Coding Agents

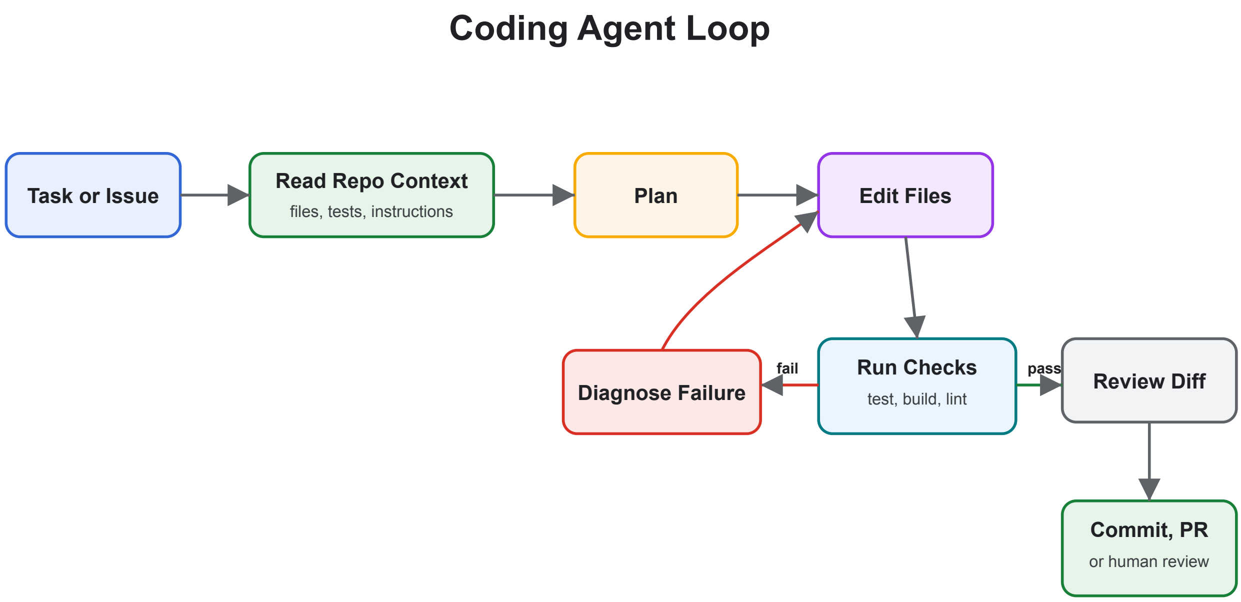
Coding agents operate inside software repositories. They read code, edit files, run commands, inspect failures, produce diffs, and often create commits or pull requests. Codex, Cursor Agent and Cloud Agent, Claude Code, OpenHands, and similar tools are examples of this architecture class.

The pattern is not “AI autocomplete.” It is a controlled development worker with repository context and execution privileges.

Examples

- Codex CLI and Codex IDE extension
- Cursor Agent, Plan Mode, and Cloud Agents
- Claude Code
- OpenHands and OpenHands GitHub

Core Loop



Surfaces

- **Local CLI:** runs near the repository and can use local tools.
- **IDE agent:** shares editor context, selected files, inline diffs, and local commands.

- **Cloud or background agent:** clones or mounts the repository in an isolated environment and returns a branch, diff, or PR.
- **CI/review agent:** reviews pull requests, comments on diffs, or proposes patches.
- **Multi-agent workspace:** runs several agents on separate branches or worktrees.

Architecture Concerns

Coding agents need unusually clear boundaries because they can change source code and run commands.

Design for:

- Repository instructions: coding standards, commands, architecture constraints, and review expectations.
- Workspace isolation: branch, worktree, container, or cloud environment per task.
- Approval policy: which commands and file edits need human approval.
- Test signal: fast checks first, then broader regression checks.
- Diff review: humans inspect changed behavior, not just final prose.
- Secret handling: no credentials in prompts, logs, or generated code.
- Dependency policy: explicit approval before adding packages or changing lockfiles.

Use When

- The task can be verified with tests, builds, type checks, screenshots, or review.
- The desired change can be described in concrete acceptance criteria.
- The agent can inspect enough repository context to follow local patterns.
- You can isolate work and review the resulting diff.

Avoid When

- The repository lacks tests or runnable checks and the change is high risk.
- The task is vague, political, or primarily product discovery.
- The agent needs broad production credentials.
- Multiple agents would edit the same files without coordination.

Operating Patterns

- Ask for a plan before large edits.
- Make the agent cite files and commands it used.
- Prefer small tasks with clear completion criteria.
- Use worktrees or branches for parallel agents.
- Require tests or type checks before commit.

- Review generated code like human code.
- Keep durable repo guidance in a project instruction file.

Failure Modes

- Plausible code that compiles but violates architecture.
- Broad refactors that mix behavior changes with formatting.
- Tests updated to match broken behavior.
- Hidden dependency changes.
- Shell commands that mutate local state unexpectedly.
- Agents fighting over the same files.
- Review fatigue when diffs are too large.

Design Rule

The coding agent should never be the only reviewer of its own code. It can propose, edit, test, and explain. A separate check, reviewer, or policy gate should decide whether the change lands.

Related Chapters

- Goals and State
- Skills
- Human Approval Gates
- Observability and Evals
- Architecture Decision Records for Agents

Architecture Decision Records for Agents

Agentic systems change quickly. Architecture Decision Records keep model, memory, tool, policy, and workflow choices explicit enough that future maintainers can understand why the system behaves the way it does.

Use ADRs when a decision affects safety, cost, reliability, user trust, or the ability to debug production behavior.

What to Record

Record decisions about:

- Model selection and fallback models
- Tool permissions and approval rules
- Memory retention and deletion
- Retrieval sources and citation policy
- Workflow retries, compensation, and escalation
- Evaluation datasets and release gates
- Observability and logging boundaries
- Human review requirements
- Self-improvement and skill-update policies

ADR Template

```
# ADR-000: Title

## Status

Proposed | Accepted | Superseded

## Context

What problem are we solving? What constraints matter?

## Decision

What did we choose?

## Consequences
```

```
What improves? What gets harder? What risks remain?

## Verification

How will we know this decision is still working?
```

Agent-Specific Additions

Add these fields when relevant:

- **Autonomy level:** advisory, proposes edits, executes after approval, or executes autonomously.
- **Tool scope:** exact tools and operations allowed.
- **State owner:** chat history, workflow state, memory store, database, or external system.
- **Failure policy:** retry, re-plan, ask, refuse, rollback, or escalate.
- **Eval gate:** tests or datasets required before release.
- **Rollback path:** how to disable or reverse the decision.

Example Decisions

- Use a durable workflow for customer-impacting tasks instead of an in-memory loop.
- Require human approval before sending outbound email.
- Store episodic memory for project events but not personal secrets.
- Use hybrid keyword plus vector retrieval for support documentation.
- Run coding agents in disposable worktrees and require `npm test` before commit.
- Keep self-improvement as reviewed skill changes, not silent prompt mutation.

Failure Modes

- Decisions live only in prompts and disappear from engineering review.
- ADRs describe the happy path but not rollback or verification.
- Model upgrades happen without recording eval impact.
- Memory or tool-scope changes ship without privacy review.
- The ADR says "human approval" but not which actions require it.

Related Chapters

- Agentic System Architecture
- Agentic RAG Systems
- Policy Enforcement

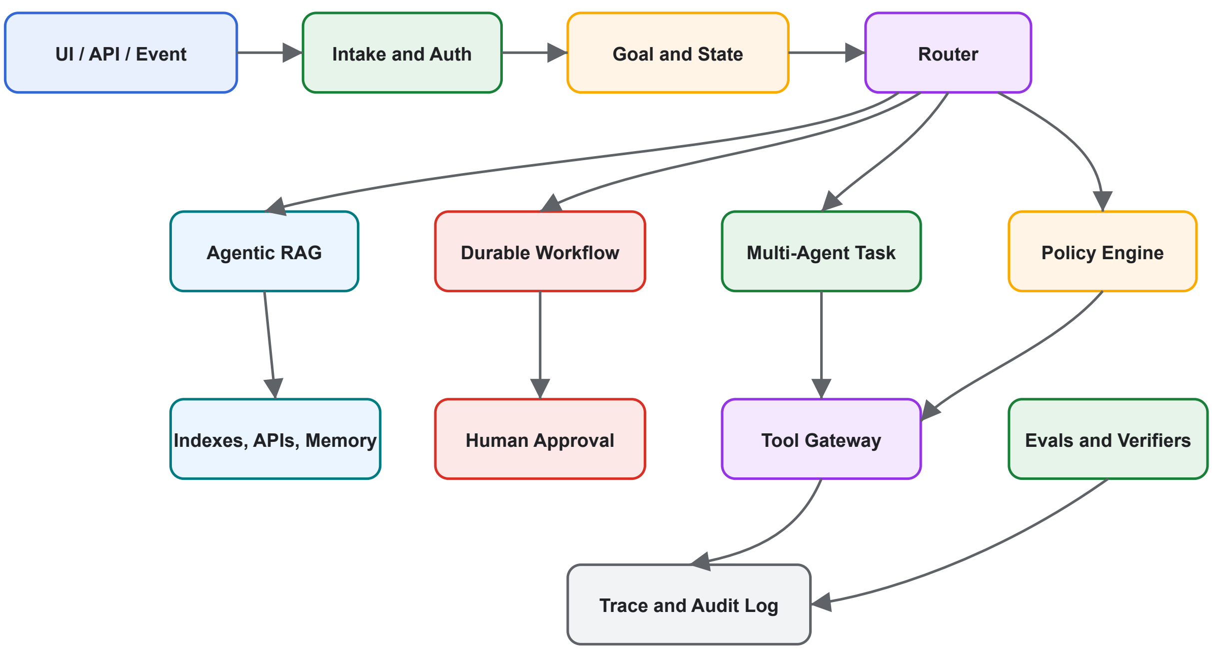
- Observability and Evals

Reference Architecture

This reference architecture combines the core patterns in the book into a production-ready agentic system. It is intentionally conservative: deterministic software owns state, policy, and side effects; model calls propose decisions inside bounded loops.

Architecture

Production Agentic System Reference Architecture



Request Flow

1. Authenticate the user or event source.
2. Create a goal record with constraints, user scope, and requested outcome.
3. Route the task to a direct answer, Agentic RAG, durable workflow, or multi-agent process.
4. Run policy checks before retrieval, tool use, or side effects.
5. Persist observations, decisions, tool calls, and outputs.
6. Verify evidence, output quality, and policy compliance.
7. Return an answer, request approval, refuse, or escalate.
8. Store traces for debugging and eval dataset improvement.

Control Points

- **Before retrieval:** enforce access control and source eligibility.
- **Before tool use:** validate schema, permission, budget, and approval needs.
- **Before final answer:** verify claims, citations, and policy.
- **Before memory write:** classify what kind of memory is being stored.
- **Before release:** run evals and regression checks.

Runtime Components

- Identity and tenant boundary
- Goal and state store
- Prompt and instruction registry
- Tool gateway
- Retrieval router
- Memory service
- Policy engine
- Approval service
- Workflow engine
- Evals service
- Trace and audit store

Minimum Production Checklist

- Explicit owner for goal and state
- Tool schemas and validation
- Human approval for high-risk actions
- Access-controlled retrieval
- Citation validation for grounded answers
- Evals for core tasks
- Traces for every run
- Budget, timeout, and cancellation controls
- Incident review path
- Rollback or disable switch

Scaling Path

Start small:

1. Single agent with tool validation.
2. Add goals and state.
3. Add retrieval and citation checks.
4. Add durable workflow for long-running tasks.
5. Add human approval gates.

6. Add eval datasets and observability.
7. Add multi-agent decomposition only when one agent becomes a bottleneck.

Related Chapters

- Agentic System Architecture
- Agentic RAG Systems
- MCP-first Tool Use
- Durable Workflows
- Policy Enforcement
- Observability and Evals

Durable Workflows

Durable workflows make agentic systems resumable and auditable by owning retries, checkpoints, approvals, compensation, and long-running state.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Durable Workflow Pattern wraps agent steps in a resumable execution model. The workflow owns ordering, retries, persisted state, approvals, and compensation; agents perform bounded work inside workflow steps.

Use When

- Work spans minutes, hours, or external approvals.
- Tool calls may fail and must be retried safely.
- You need auditability, resumability, and operational control.

Avoid When

- The task is a short interactive chat response.
- State does not need to survive process restarts.
- The workflow engine would add more complexity than the task deserves.

Architecture

```
flowchart TD
  E[Event] --> W[Workflow State]
  W --> S1[Step: Agent or Tool]
  S1 --> C{Checkpoint}
  C --> S2[Next Step]
  S2 --> H{Approval Needed?}
  H -->|yes| A[Human Approval]
  H -->|no| D[Done]
  A --> S2
```

Implementation Notes

- Persist state after every externally visible action.
- Make steps idempotent or attach idempotency keys to tool calls.
- Separate orchestration state from model conversation state.
- Model calls should be retryable only when repeating them cannot duplicate side effects.

Failure Modes

- Retrying side-effectful steps without idempotency.
- Losing human approval state after deployment or restart.
- Workflows that hide agent uncertainty behind a successful task status.
- No compensation path for partially completed external actions.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `durable-workflow-pattern/` folder from this repository.

Related Patterns

- Goals and State
- Self-Healing Workflow
- Human-in-the-Loop Approval

Observability and Evals

Observability records what happened. Evals decide whether behavior is good enough.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Observability and Evals Pattern makes agent behavior inspectable and testable. It captures traces, tool calls, prompts, model outputs, costs, latencies, and evaluation results so teams can debug and improve systems over time.

Use When

- Agent decisions affect users, money, data, or external systems.
- You need regression tests for prompts, tools, routing, or workflows.
- Failures are hard to reproduce from final answers alone.

Avoid When

- You cannot store traces safely because of privacy or regulatory constraints.
- The prototype is throwaway and has no operational users.
- You only log final answers and call that observability.

Architecture

```
flowchart LR
  R[Run] --> T[Trace]
  R --> M[Metrics]
  R --> E[Eval Dataset]
  T --> D[Debugging]
  M --> O[Operations]
  E --> Q[Quality Gates]
```

Implementation Notes

- Trace at the level of run, loop iteration, model call, tool call, workflow step, and evaluator result.
- Store enough input/output detail to reproduce failures, with redaction for sensitive data.
- Maintain golden datasets for routing, structured outputs, tool plans, and final answers.
- Treat eval failures as release blockers for production agents.

Failure Modes

- Logs that omit the prompt, tool input, or model configuration.
- Evals that only check happy paths.
- Metrics without trace IDs, making incidents hard to investigate.
- Storing sensitive data without retention or redaction rules.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

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Download

- Download source bundle
- Open source folder

The download bundle contains the current `observability-and-evals-pattern/` folder from this repository.

Related Patterns

- Evaluator-Optimizer
- Mastra Runtime
- Compliance/Policy Enforcer

Policy Enforcement

Policy enforcement constrains what the agent may say or do through permissions, data-access rules, business rules, safety rules, and escalation.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Policy enforcement constrains what the agent may say or do through permissions, data-access rules, business rules, safety rules, and escalation.

Use When

- Actions must be checked before execution.
- The agent handles regulated, private, or business-critical data.
- Policy decisions must be auditable.

Avoid When

- Policy is only written as prompt text with no runtime enforcement.
- The system cannot identify the actor, resource, action, and context.
- Exceptions are silent or unreviewed.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `compliance-policy-enforcer-agent/` folder from this repository.

Event-Triggered Agents

Event-triggered agents run in response to webhooks, queues, schedules, or domain events.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

Event-triggered agents run in response to webhooks, queues, schedules, or domain events.

Use When

- A domain event should start bounded agent work.
- The task can be idempotent and observable.
- No human may be watching the interaction live.

Avoid When

- The event payload lacks enough context to act safely.
- Duplicate delivery could cause duplicate side effects.
- Failures cannot be retried or dead-lettered.

Implementation Notes

- Keep the pattern boundary explicit: inputs, state, side effects, and outputs should be visible.
- Validate model-produced decisions before they affect tools, users, or durable state.
- Emit enough trace data to debug failures after the run.

Failure Modes

- The pattern is applied where a simpler deterministic workflow would be better.
- State, tool calls, or model decisions are not observable enough to debug.
- The system lacks clear stop, retry, or escalation behavior.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `event-triggered-agent-pattern/` folder from this repository.

Mastra Runtime

Mastra is a TypeScript runtime pattern for applications that need agents, workflows, tools, memory, evals, and observability in one framework.

> Source and downloads
>
> - Repository source
> - Download code bundle

Intent

The Mastra Runtime Pattern uses Mastra as a TypeScript runtime for production agent applications. Mastra gives agents, workflows, tools, memory, evals, and observability a shared application structure.

Use When

- You are building a TypeScript or Node-based agent product.
- You need agents and deterministic workflows in the same runtime.
- You want memory, tools, evals, and tracing to be first-class concerns.

Avoid When

- You only need a small script or single model call.
- Your team is committed to a Python-first agent stack.
- You cannot accept framework conventions around project structure and deployment.

Architecture

```
flowchart TD
  App[Application] --> Agent[Mastra Agent]
  App --> Workflow[Mastra Workflow]
  Agent --> Tools[Tools]
  Agent --> Memory[Memory]
  Workflow --> Agent
  Workflow --> Evals[Evals]
  App --> Obs[Observability]
```

Implementation Notes

- Use agents for open-ended decisions where the next step is not known upfront.
- Use workflows for predetermined control flow, state transitions, retries, and production orchestration.
- Keep tools typed and independently testable.
- Capture traces and evals from the beginning rather than adding them after failures.

Failure Modes

- Treating the framework as the architecture instead of modeling goals, state, and failure modes.
- Putting deterministic workflow logic inside prompts.
- Creating tools with vague descriptions and unvalidated inputs.
- Shipping without eval datasets or trace review.

Code Walkthrough

Read the excerpt as the smallest executable expression of the pattern. The surrounding chapter explains the design constraints; the code shows where those constraints become concrete interfaces, state, validation, or control flow.

Source Code

This pattern currently has no dedicated code excerpt. Use the source and download links below for the full pattern folder.

Download

- Download source bundle
- Open source folder

The download bundle contains the current `mastra-runtime-pattern/` folder from this repository.

Related Patterns

- Durable Workflow
- Observability and Evals
- Agent Loop

Deprecated / Historical Patterns

Deprecated patterns are preserved in the repository but removed from the active learning path.

They are either speculative, too broad, duplicated by stronger chapters, or currently too thin to stand as active patterns.

- Agent Marketplace: speculative compared with A2A, routing, and orchestration.
- Agent Swarm: folded into Parallel Agents, search, and optimization.
- Hybrid Agent: too broad to teach as a standalone pattern.
- Meta-Cognitive Agent: folded into Reflection, Evaluator-Optimizer, and Self-Improvement.
- Recursive Agent: folded into Planning, Goals and State, and Task Delegation.
- Distributed Agent: replaced by A2A, durable workflows, and protocol-first composition.
- API Integration Copilot: archived as an applied placeholder.
- Data Pipeline Orchestrator Agent: archived as an applied placeholder.
- Multi-Modal Tool-Using Agent: archived until it has a developed multimodal implementation.

Source archive: `deprecated`